

Autoencoder-based generation of individuals in population-based metaheuristics



February 19th, 2020

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OUTLINE

- ▶ Autoencoders & genetic algorithms
- ▶ Proposed operator
- ▶ Results & analyses
- ▶ Conclusions & perspectives

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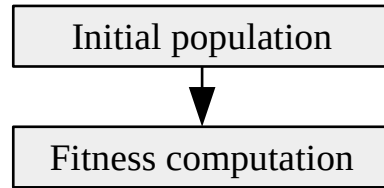
Autoencoders & genetic algorithms

└ Main elements of a genetic algorithm

Initial population

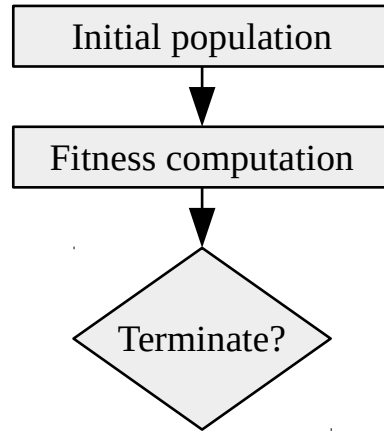
Autoencoders & genetic algorithms

└ Main elements of a genetic algorithm



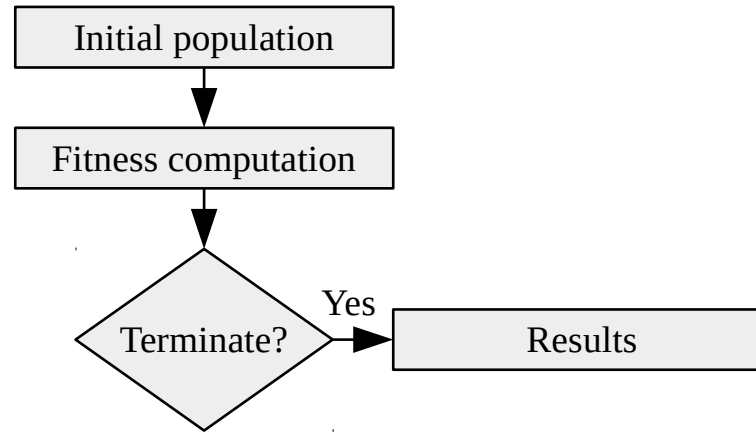
Autoencoders & genetic algorithms

└ Main elements of a genetic algorithm



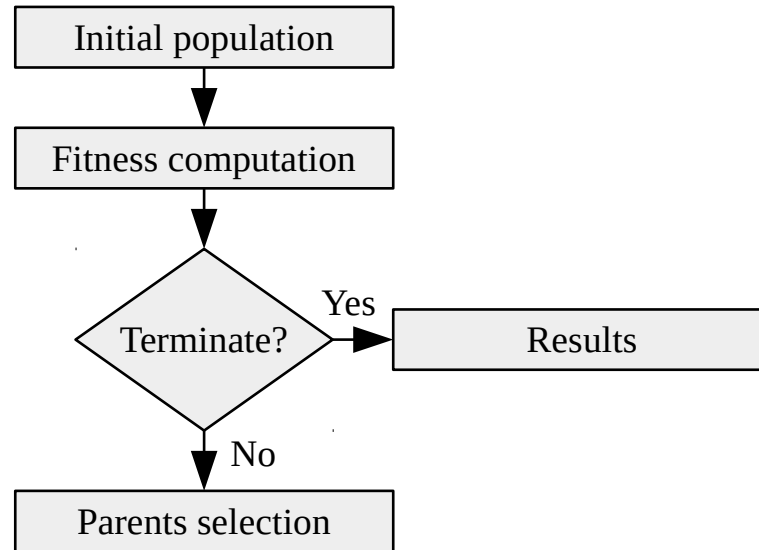
Autoencoders & genetic algorithms

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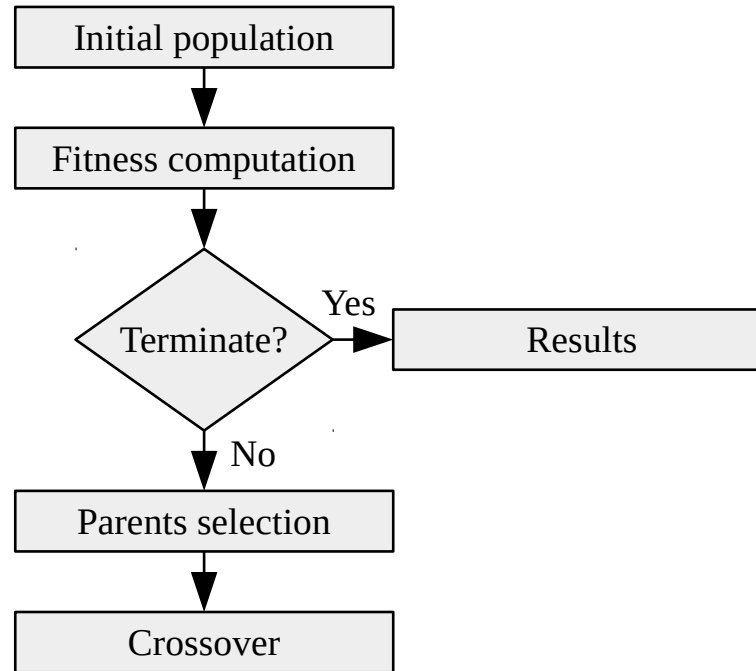
Autoencoders & genetic algorithms

└ Main elements of a genetic algorithm



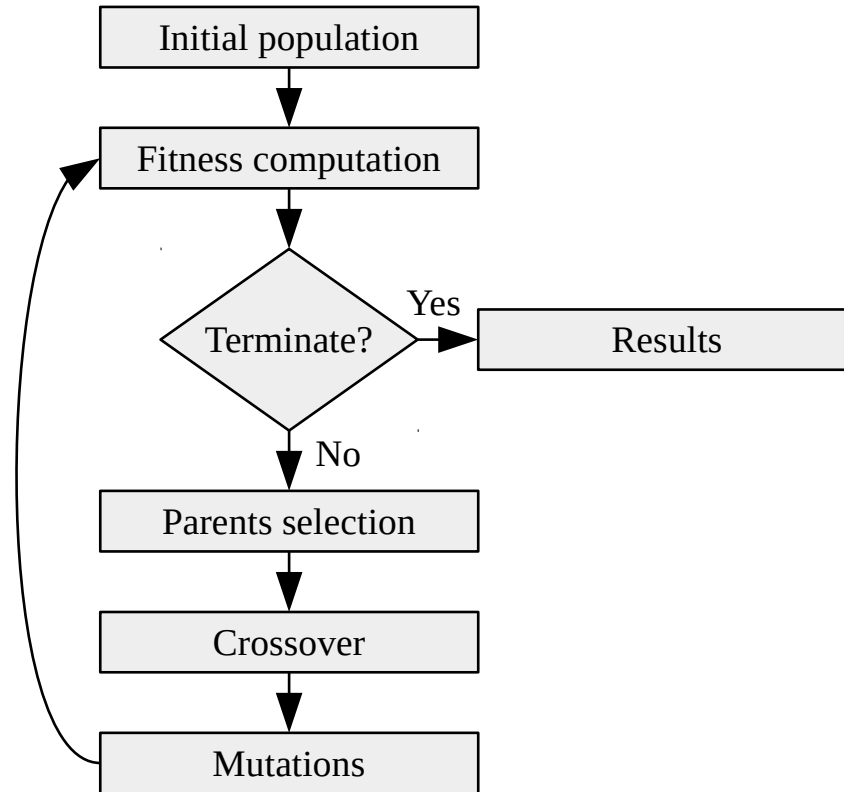
Autoencoders & genetic algorithms

└ Main elements of a genetic algorithm



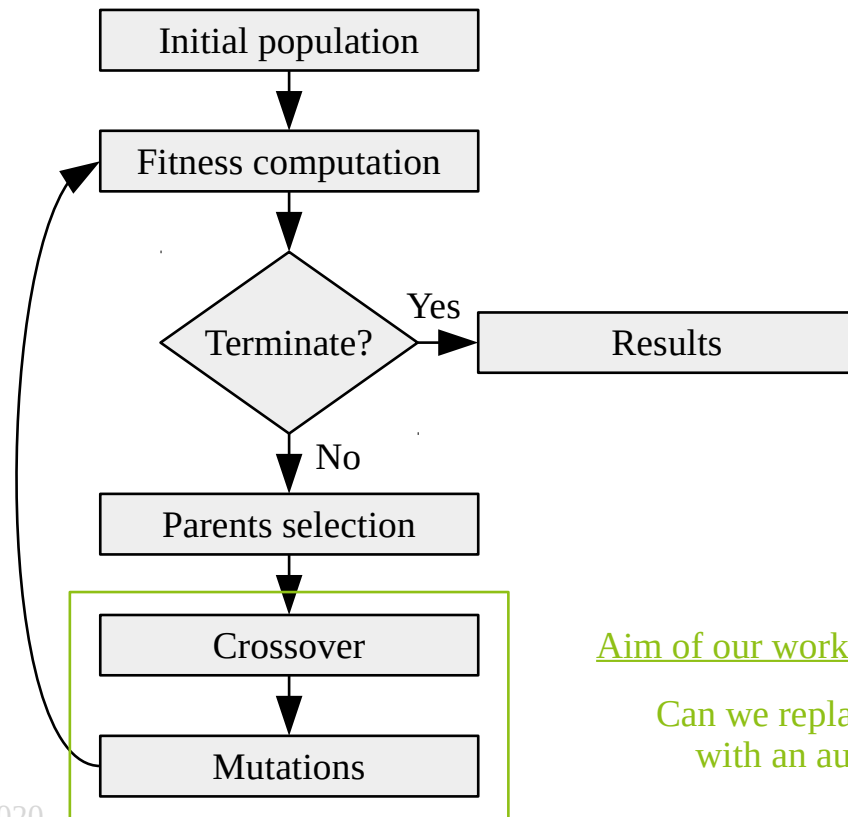
Autoencoders & genetic algorithms

└ Main elements of a genetic algorithm



Autoencoders & genetic algorithms

└ Main elements of a genetic algorithm

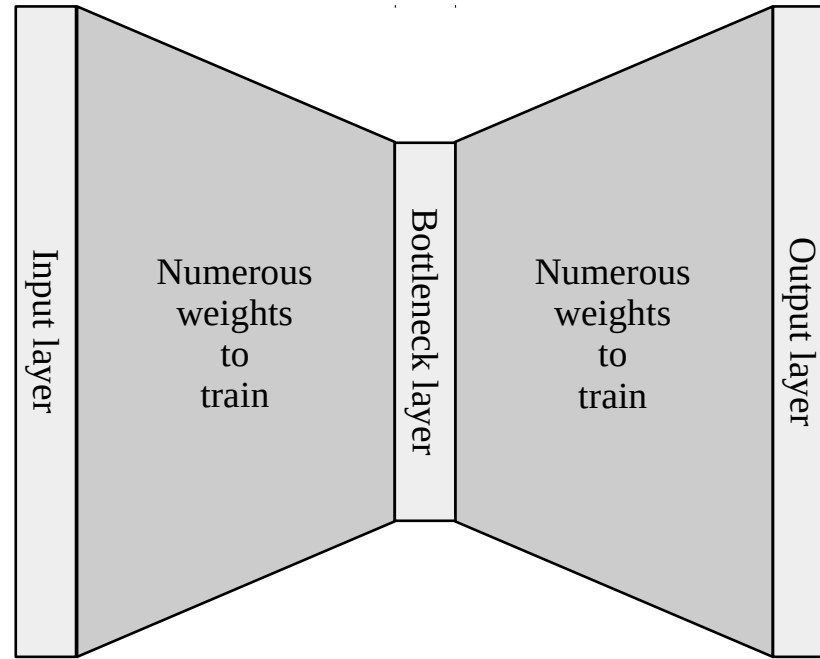


Aim of our work:

Can we replace these parts
with an autoencoder?

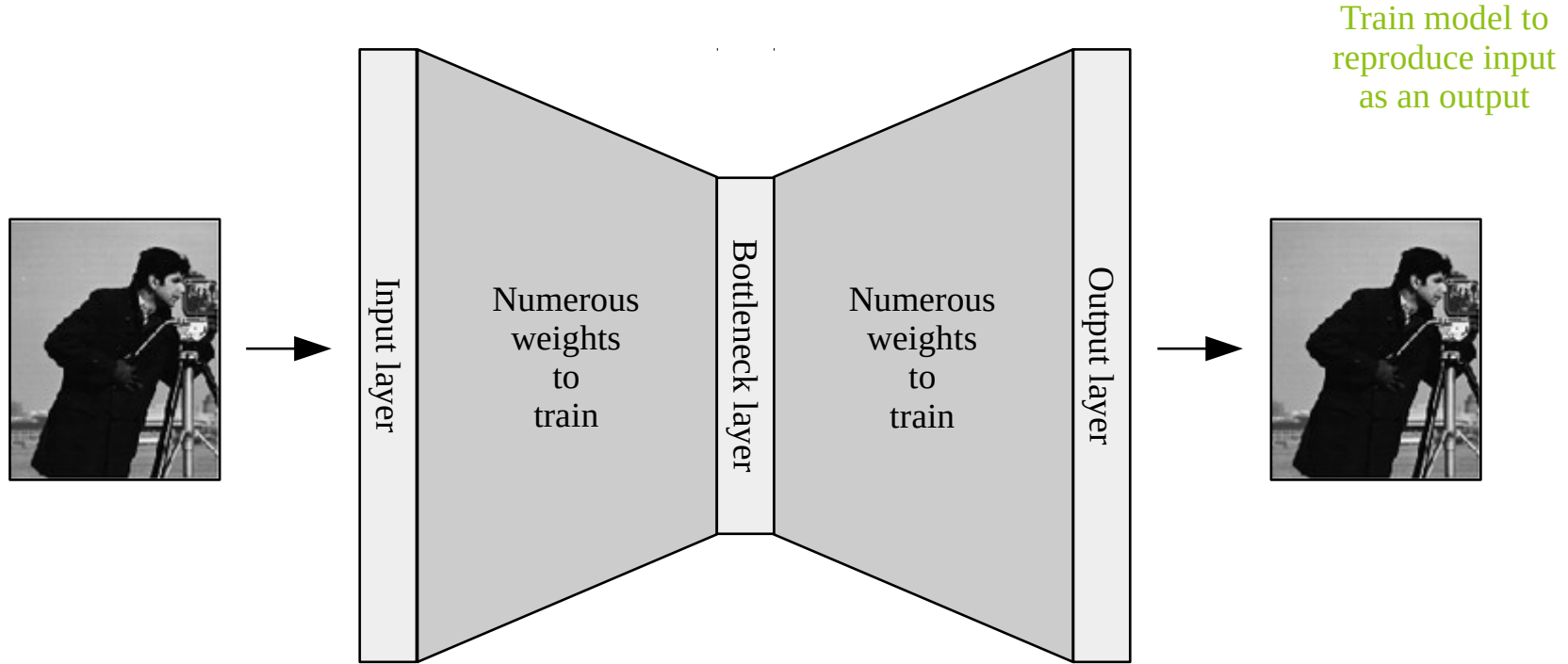
Autoencoders & genetic algorithms

└ What is an autoencoder?

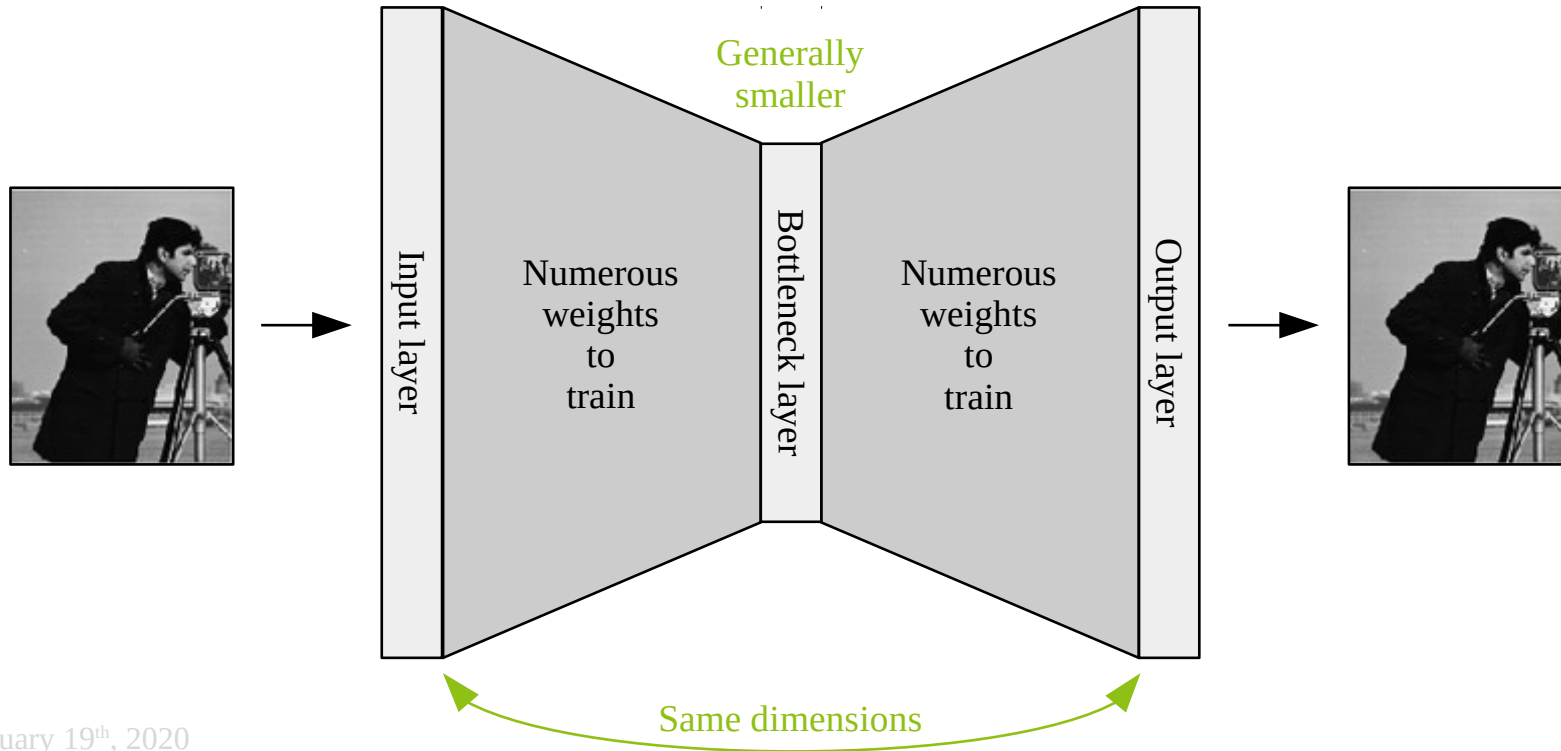


Autoencoders & genetic algorithms

└ What is an autoencoder?

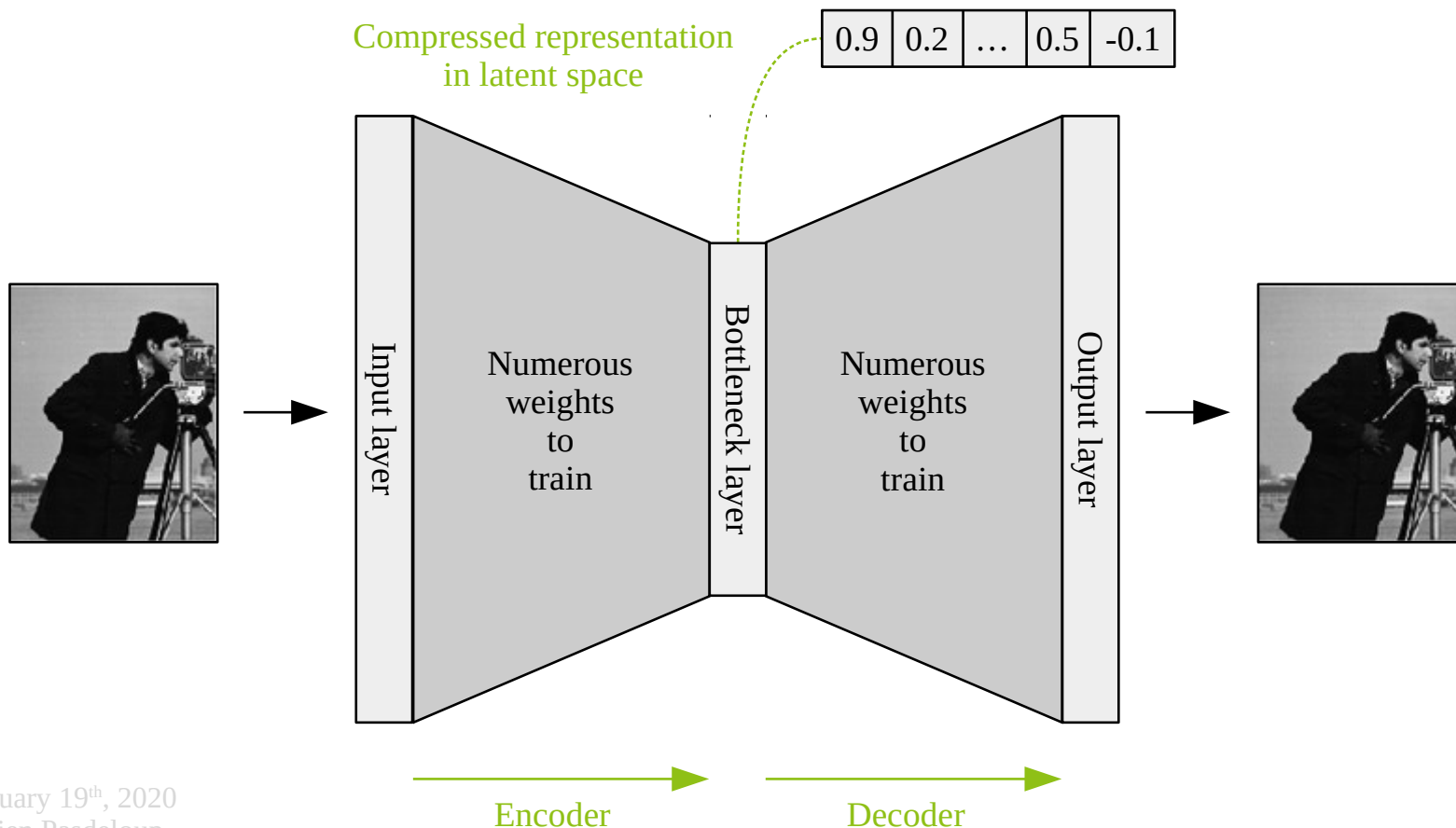


Autoencoders & genetic algorithms



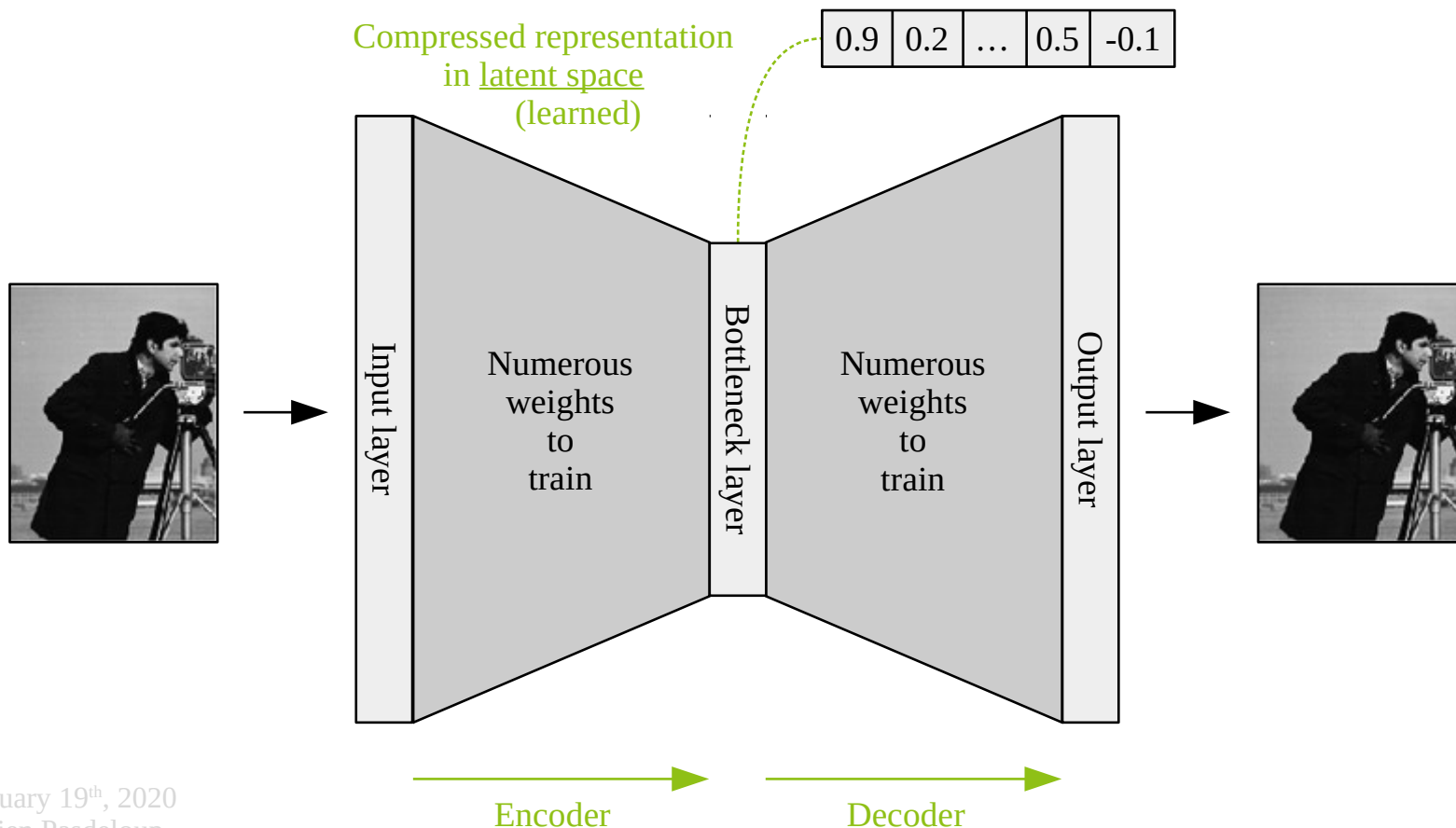
Autoencoders & genetic algorithms

└ What is an autoencoder?



Autoencoders & genetic algorithms

└ What is an autoencoder?

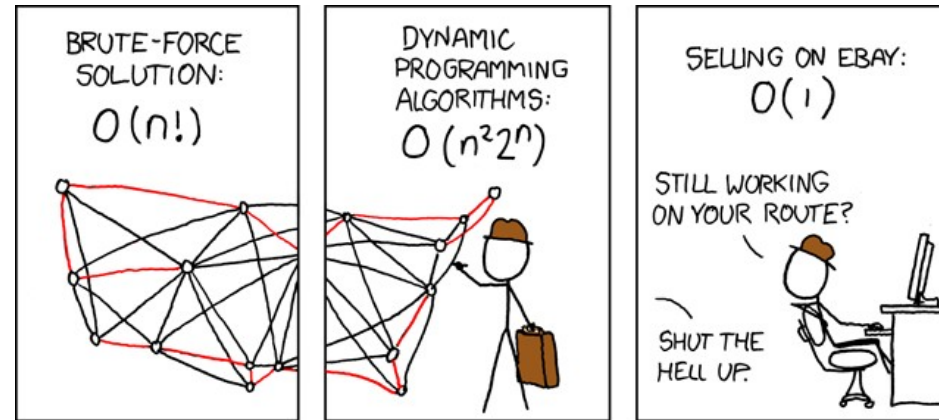


OUTLINE

- ▶ Autoencoders & genetic algorithms
- ▶ **Proposed operator**
- ▶ Results & analyses
- ▶ Conclusions & perspectives

Proposed operator

└ The traveling salesman problem (TSP)



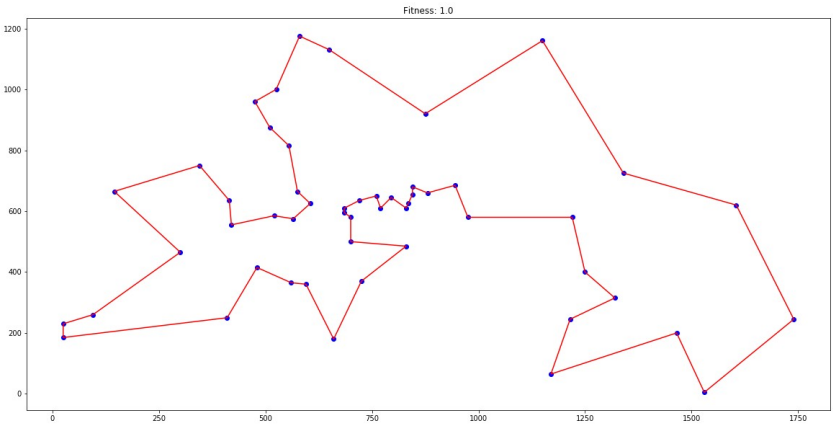
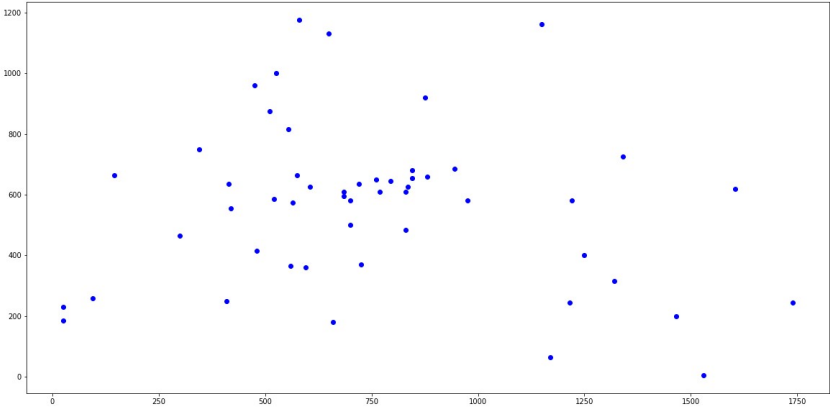
<https://xkcd.com/399/>

Proposed operator

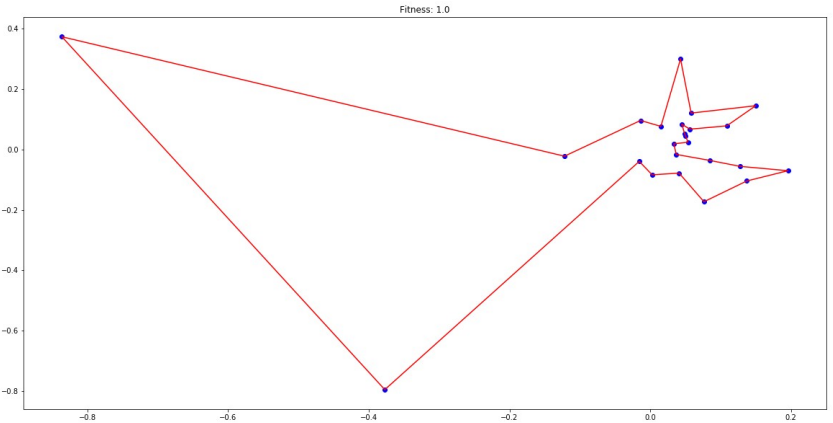
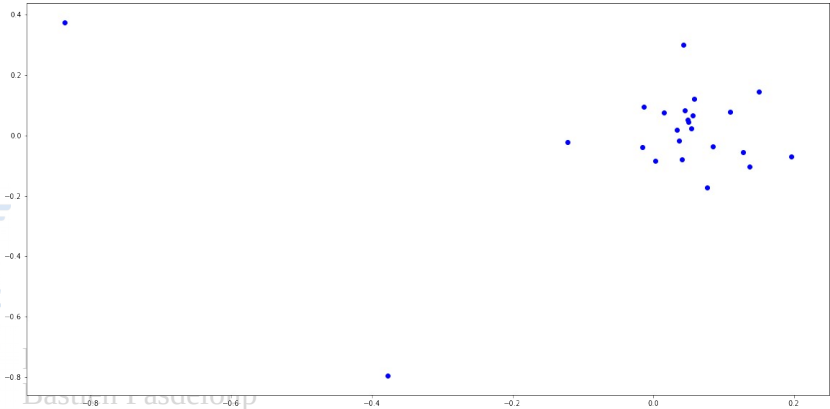
└ TSPLIB library



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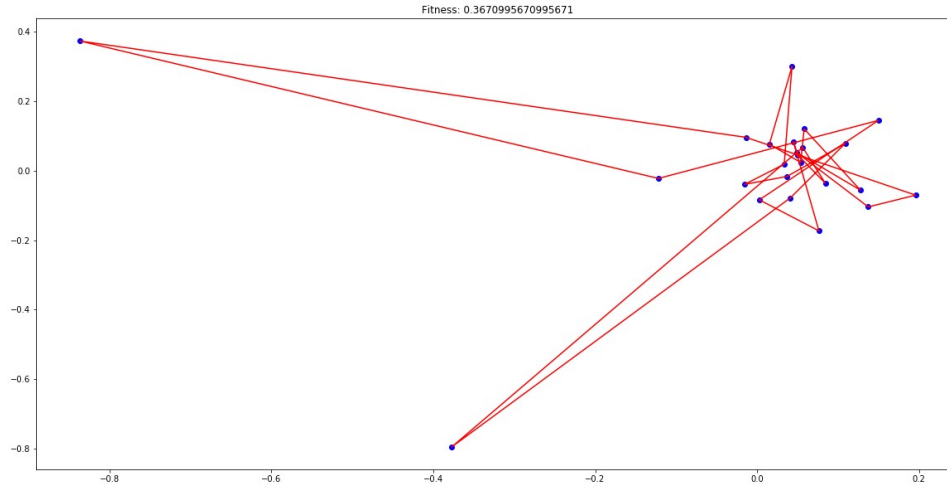


Gr24



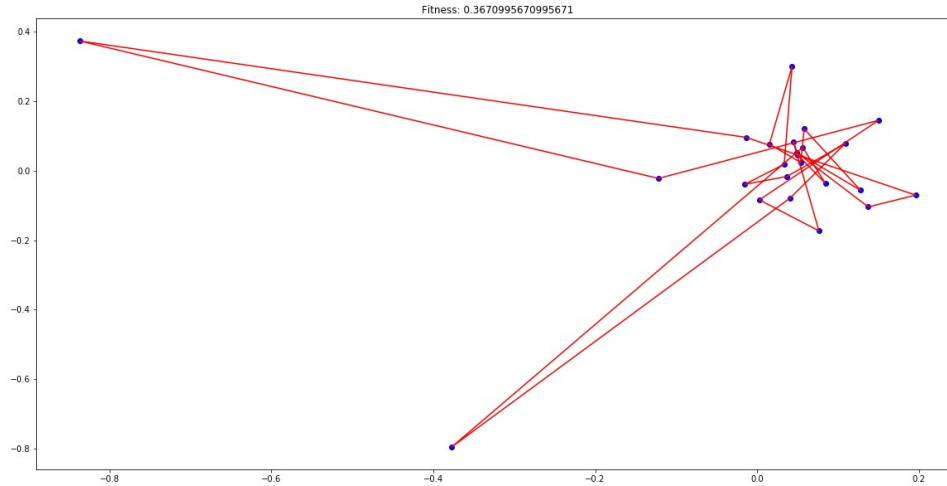
Proposed operator

└ One-hot encoding of a solution to the TSP

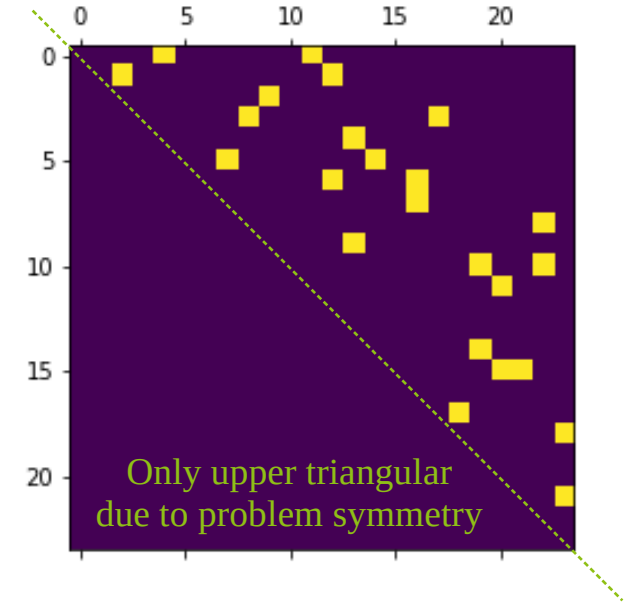


Proposed operator

└ One-hot encoding of a solution to the TSP

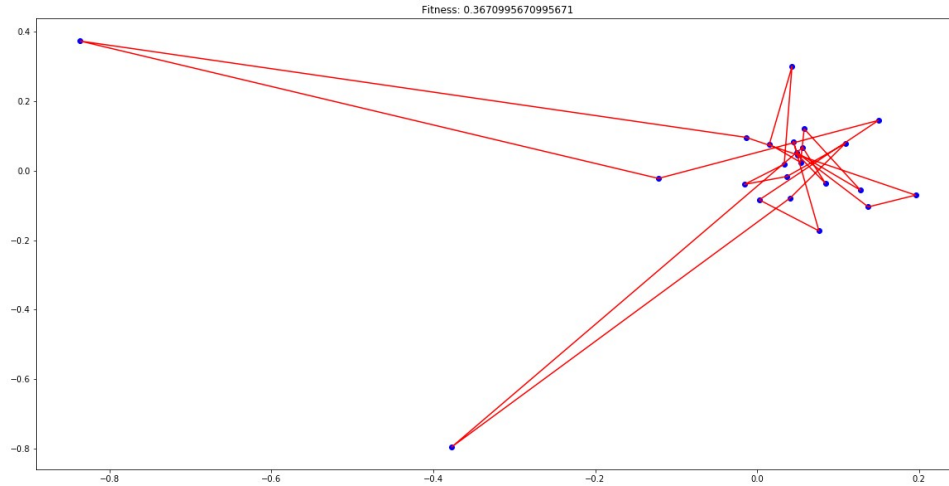


Represent as
adjacency matrix

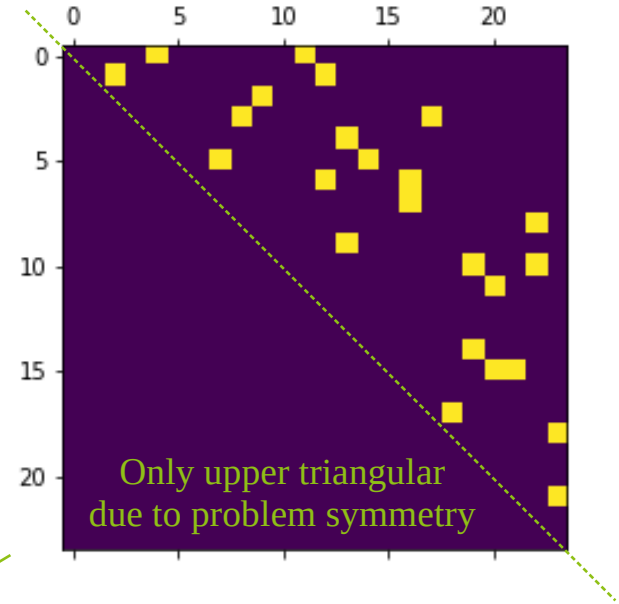


Proposed operator

└ One-hot encoding of a solution to the TSP



Represent as
adjacency matrix

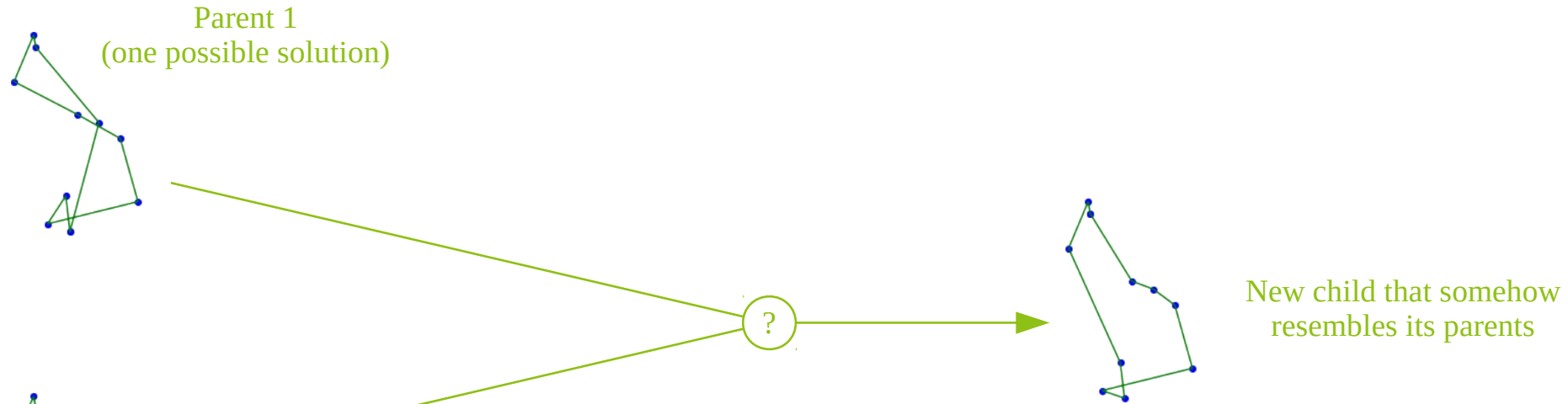


Vectorize
the matrix



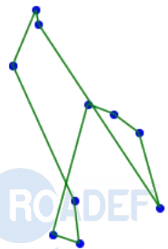
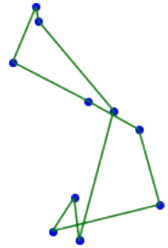
Proposed operator

└ Plugging autoencoders inside population-based metaheuristics



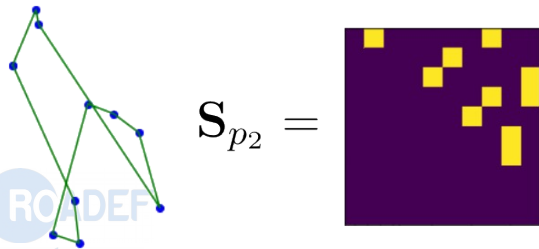
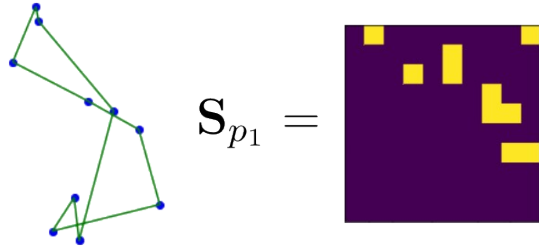
Proposed operator

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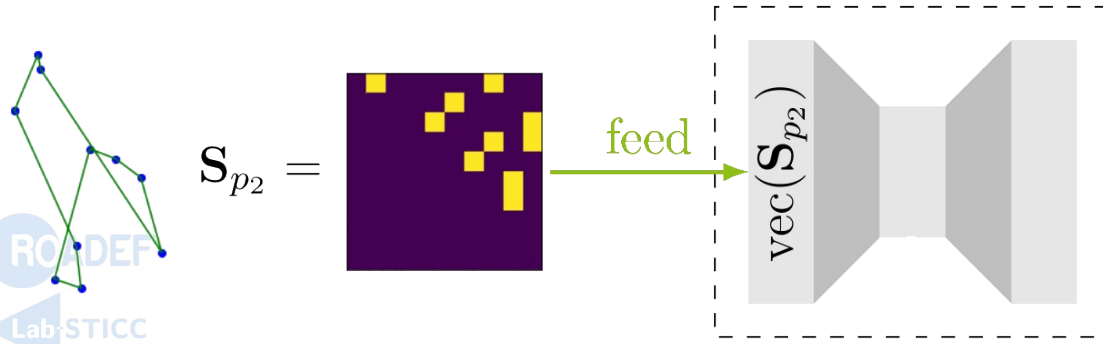
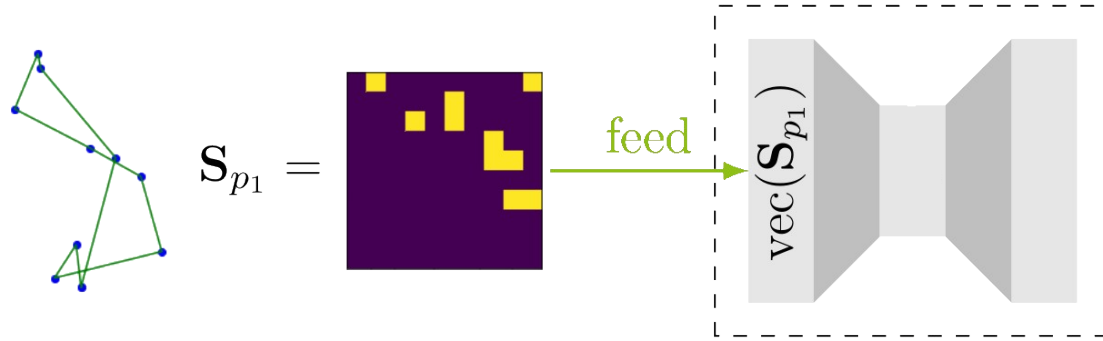
Proposed operator

└ Plugging autoencoders inside population-based metaheuristics



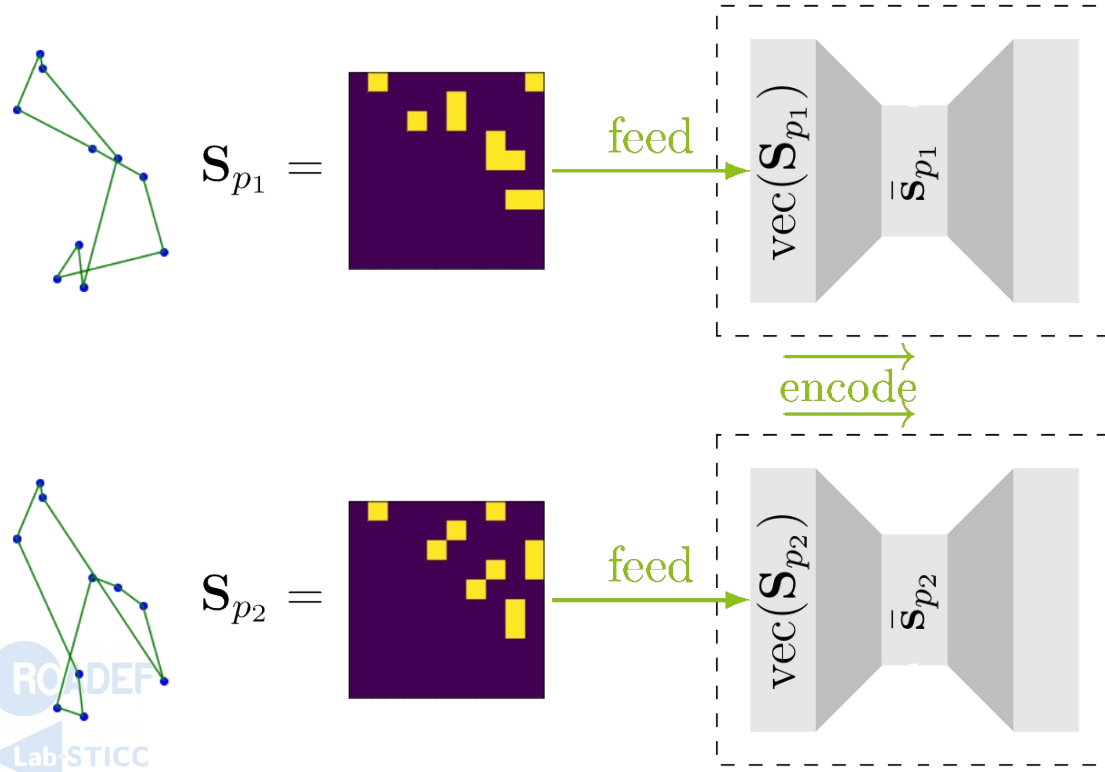
Proposed operator

└ Plugging autoencoders inside population-based metaheuristics



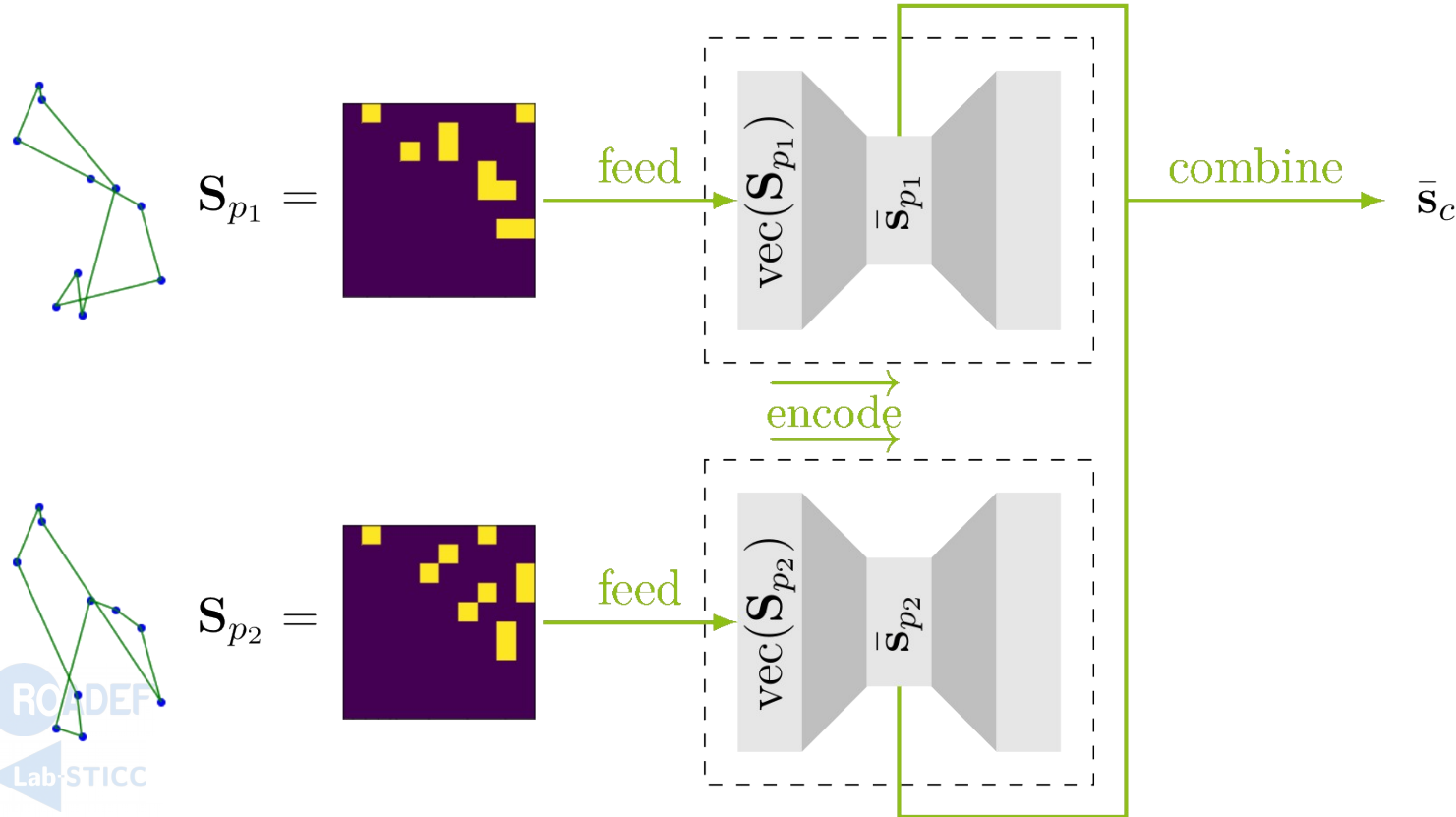
Proposed operator

└ Plugging autoencoders inside population-based metaheuristics



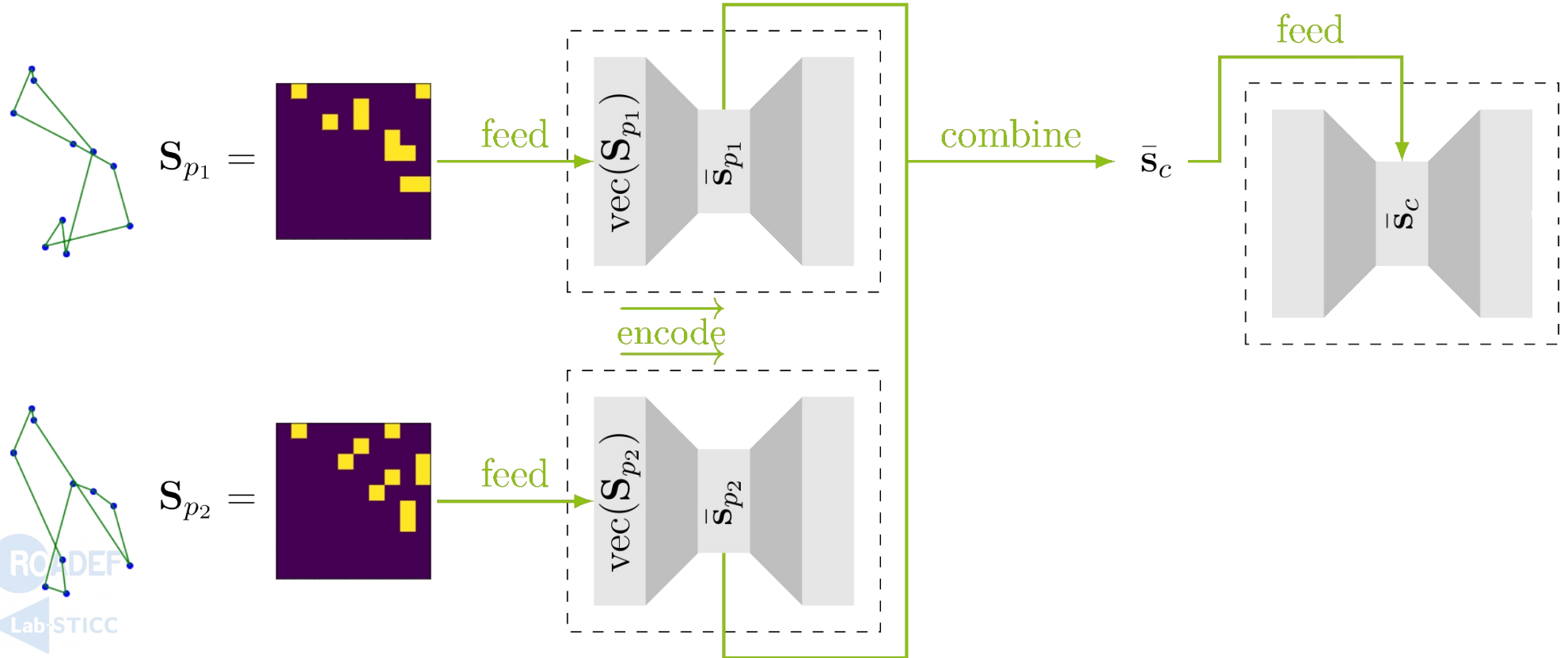
Proposed operator

└ Plugging autoencoders inside population-based metaheuristics



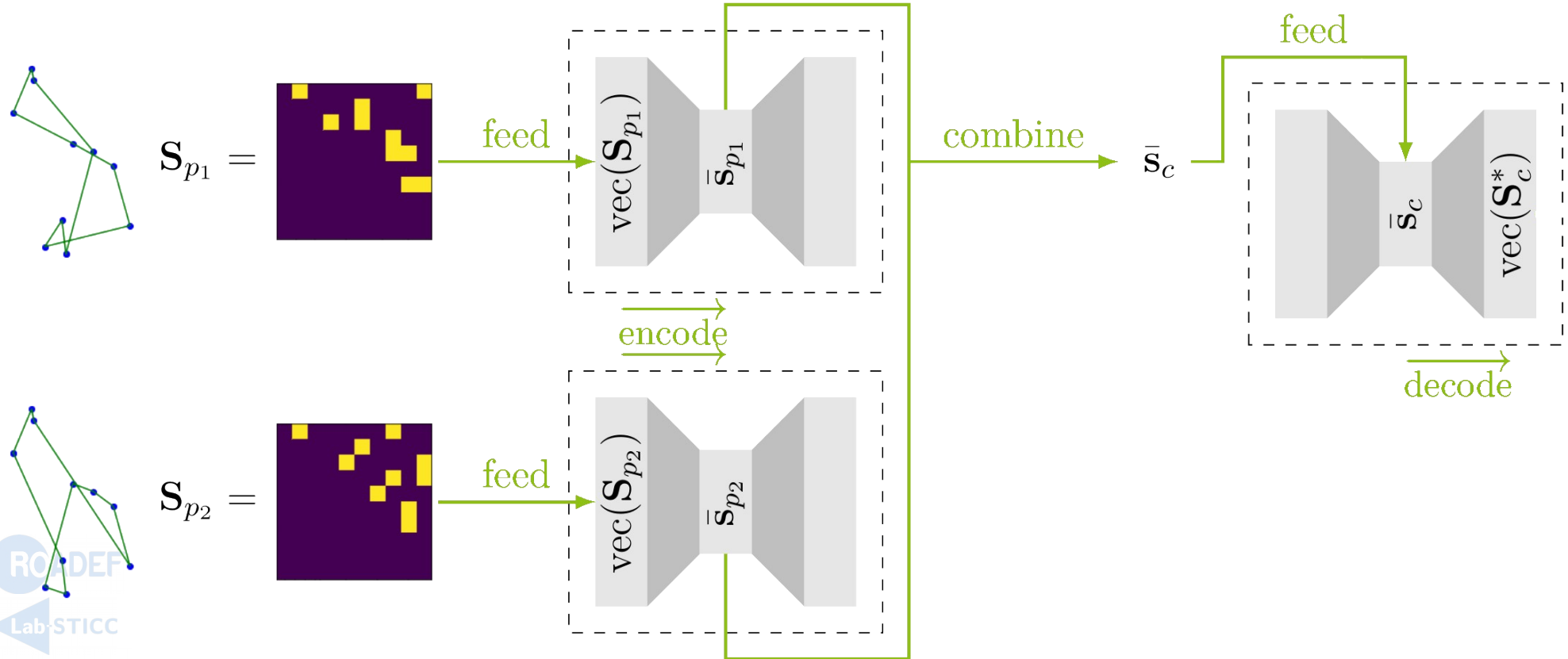
Proposed operator

└ Plugging autoencoders inside population-based metaheuristics



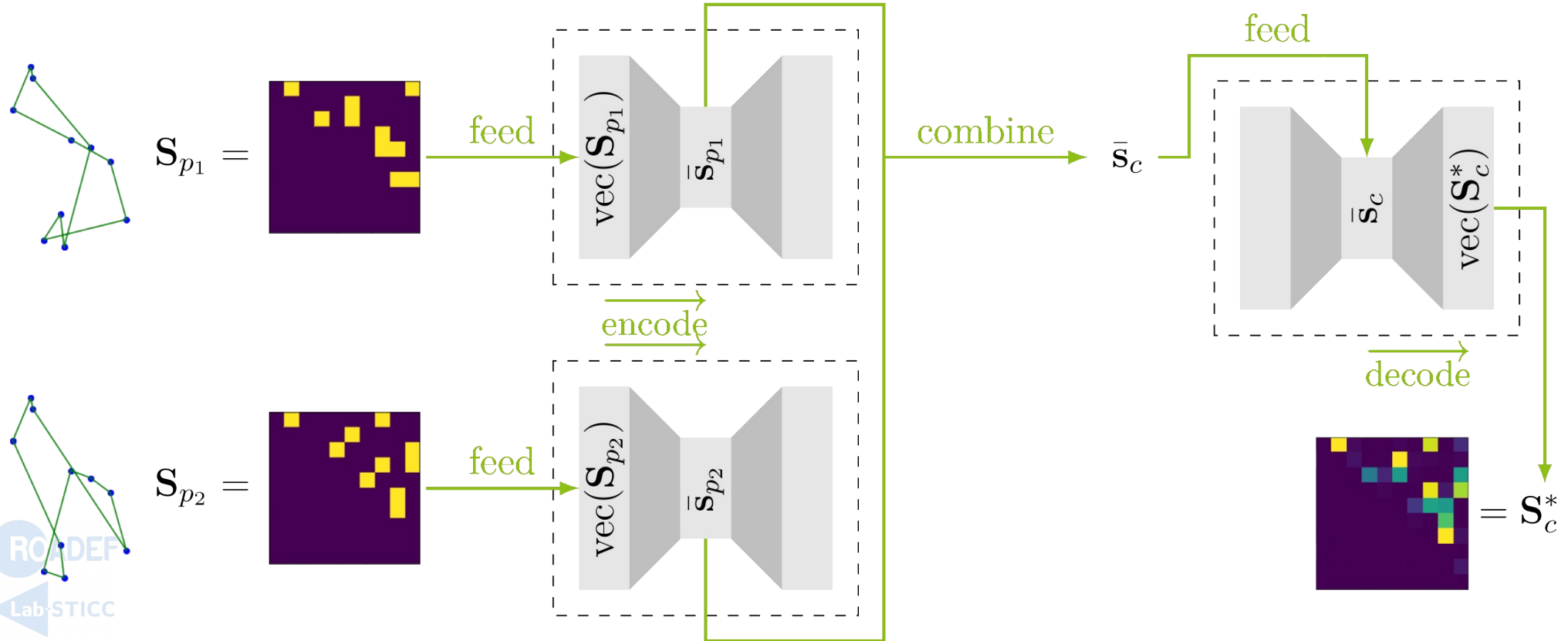
Proposed operator

└ Plugging autoencoders inside population-based metaheuristics



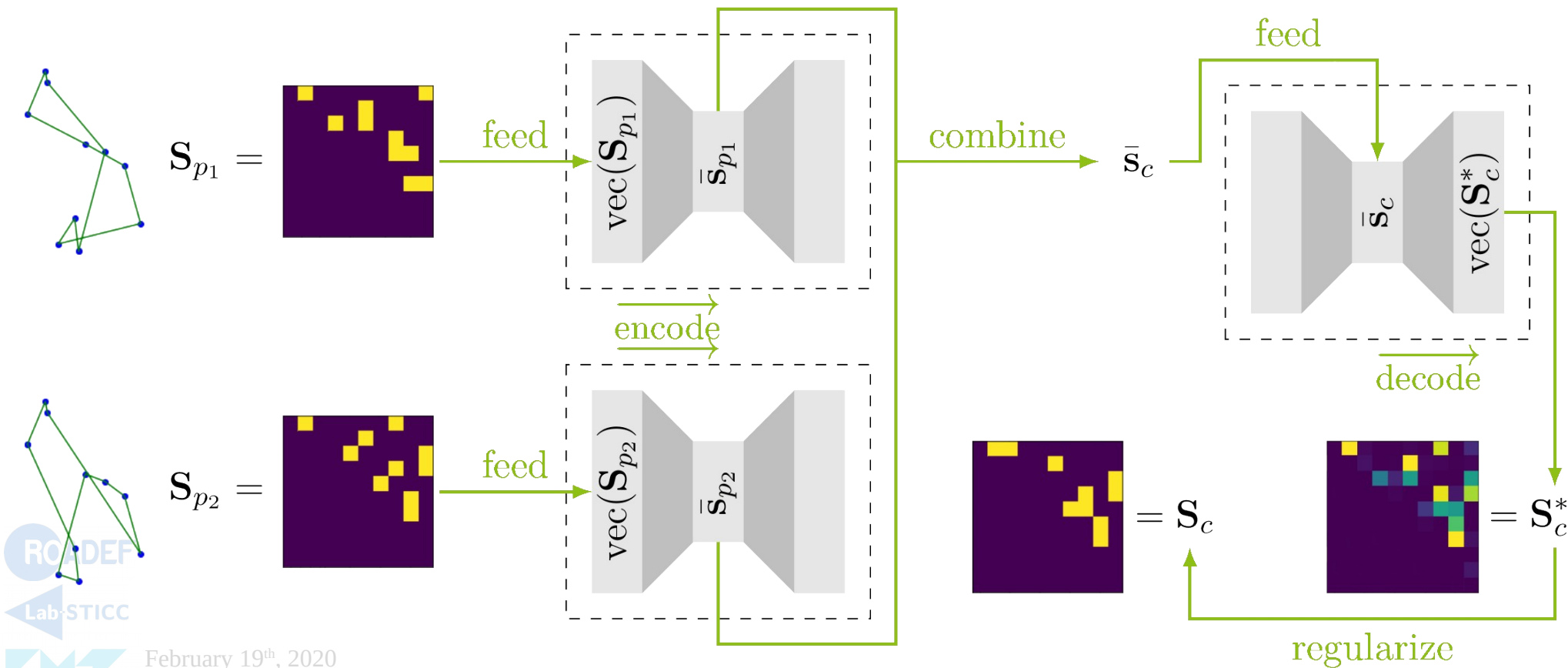
Proposed operator

└ Plugging autoencoders inside population-based metaheuristics



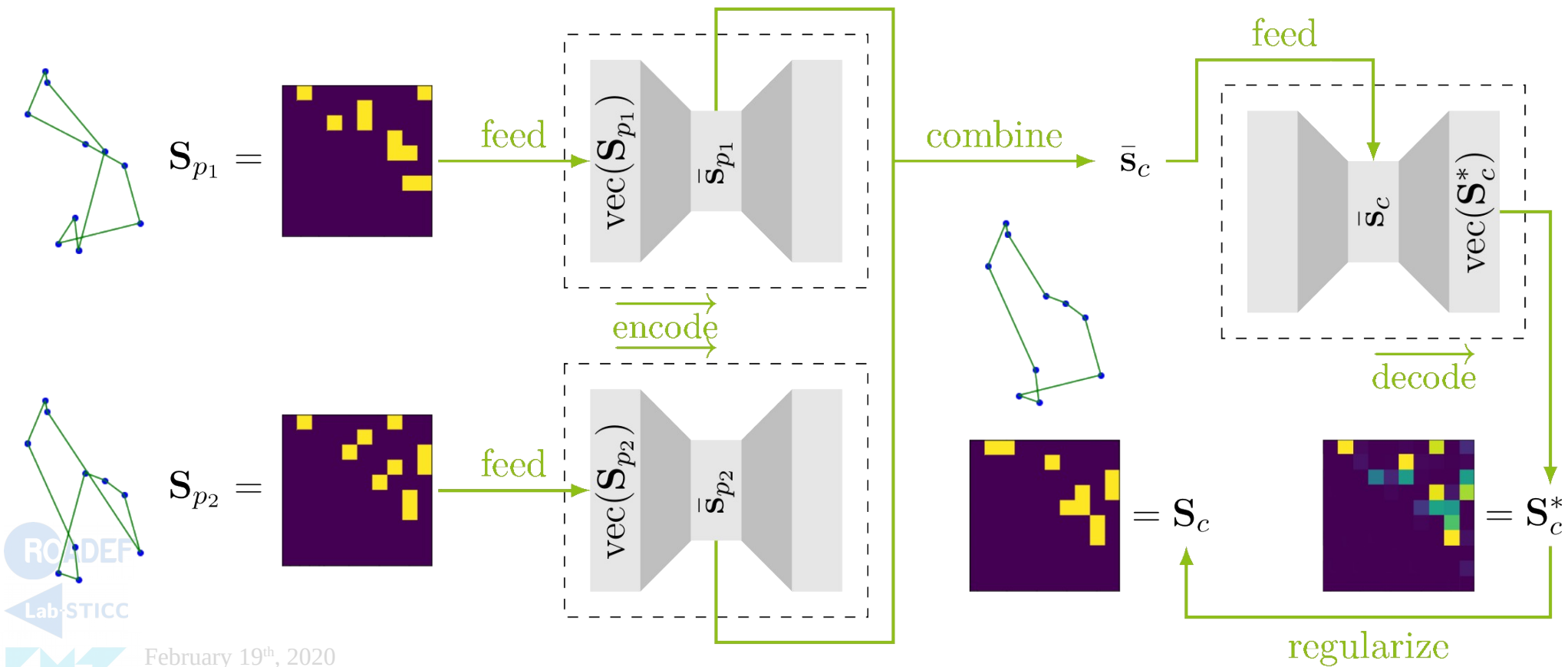
Proposed operator

└ Plugging autoencoders inside population-based metaheuristics



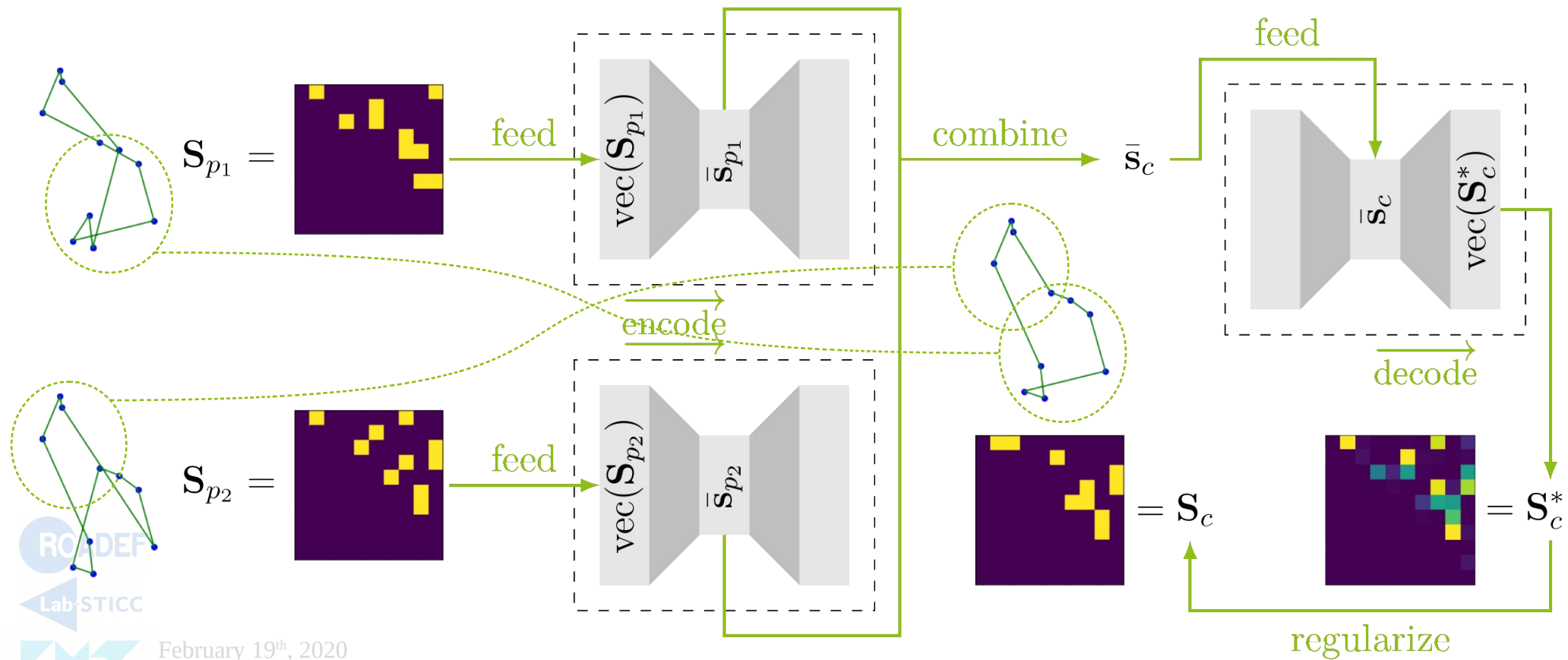
Proposed operator

└ Plugging autoencoders inside population-based metaheuristics



Proposed operator

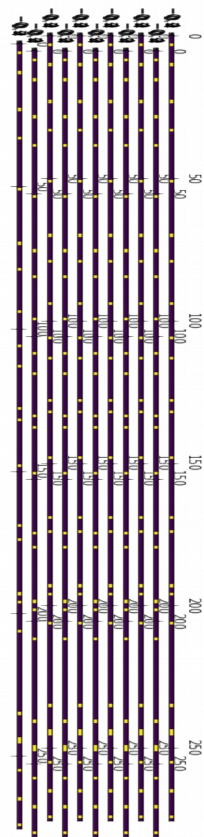
└ Plugging autoencoders inside population-based metaheuristics



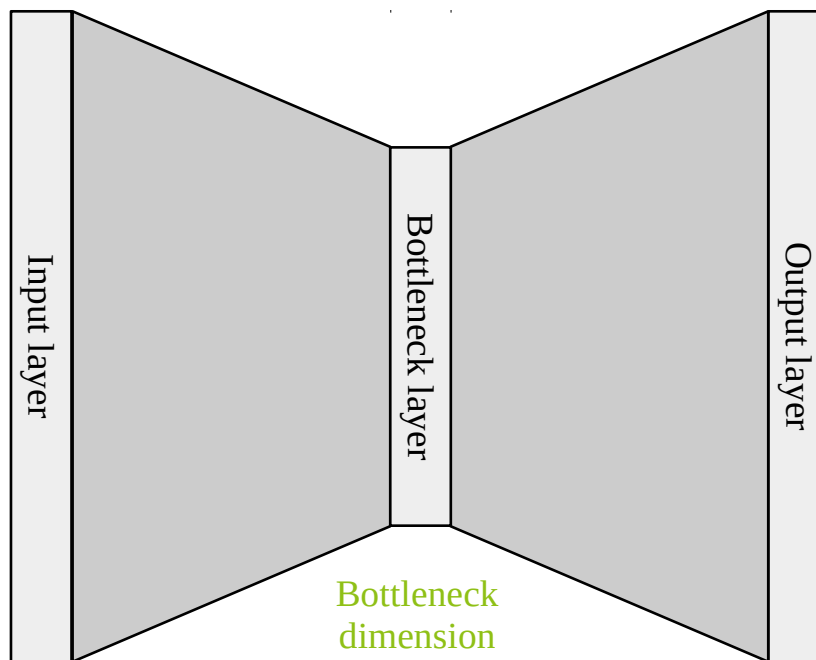
Proposed operator

└ Parameters of the approach (training the model)

Training set /
Validation set
sizes

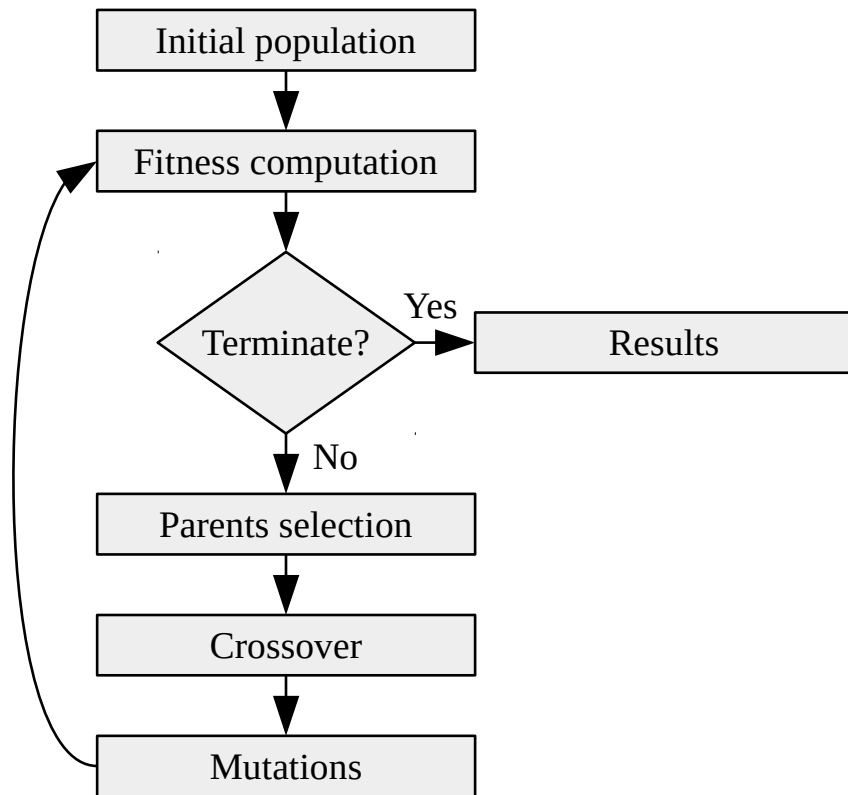


Batch size / Optimizer / Learning rate / Epochs



Proposed operator

Parameters of the approach (genetic algorithm)

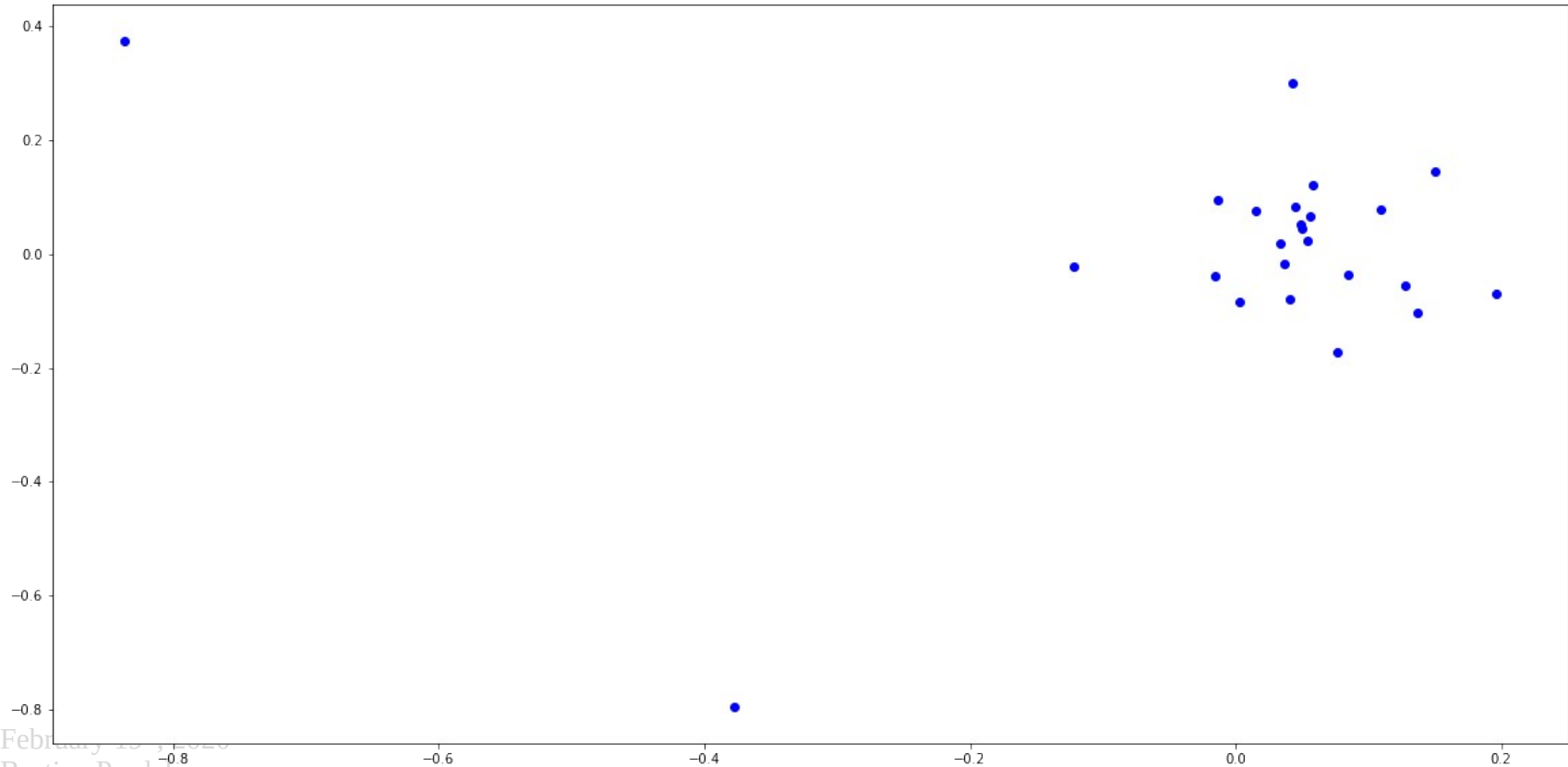


- Nb parents per child
- Parents selection method
- Population size
- Elitism
- Random insertions
- Iterations
- ...

OUTLINE

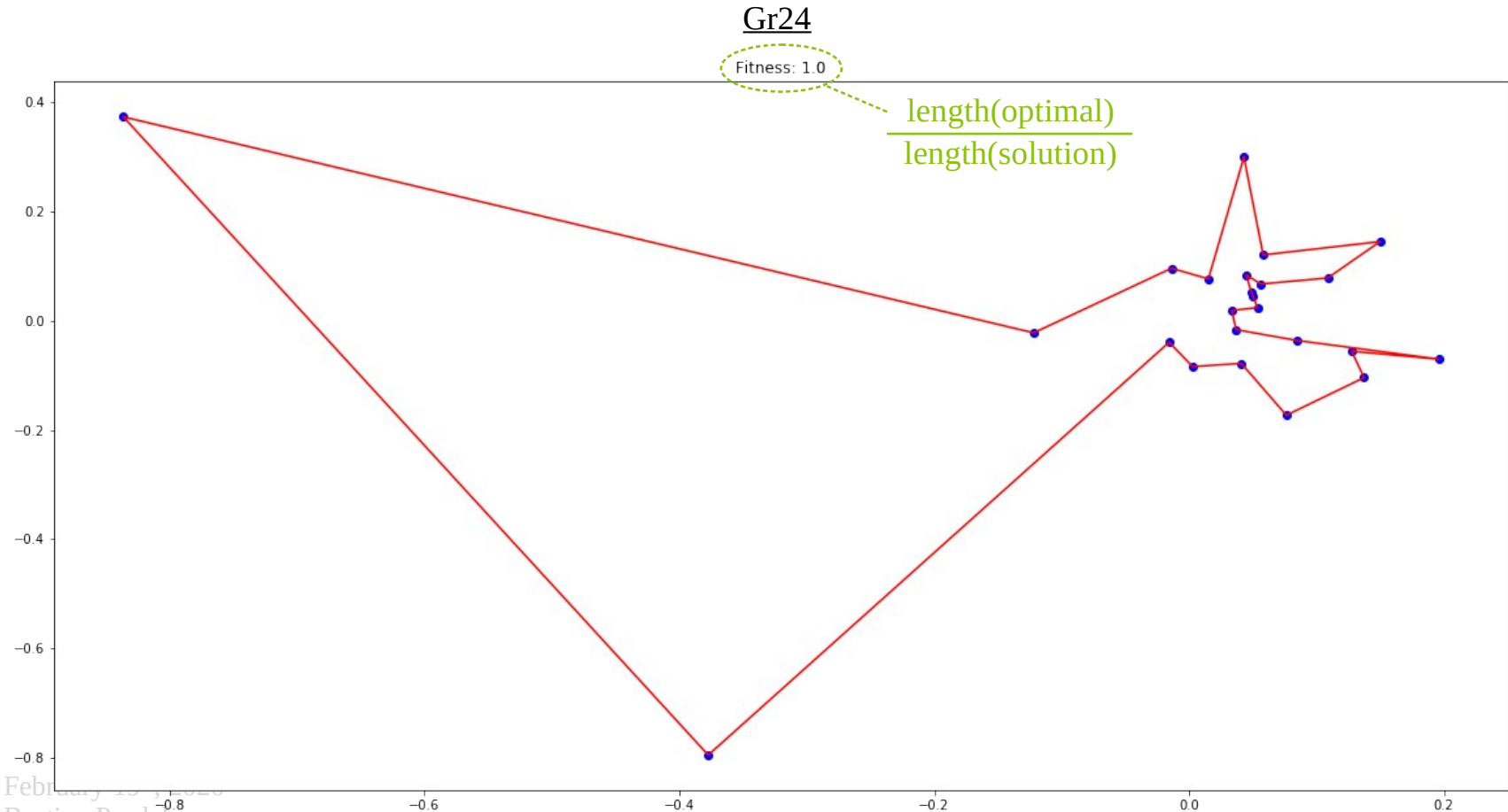
- ▶ Autoencoders & genetic algorithms
- ▶ Proposed operator
- ▶ **Results & analyses**
- ▶ Conclusions & perspectives

Obtained results

Gr24

Results & analyses

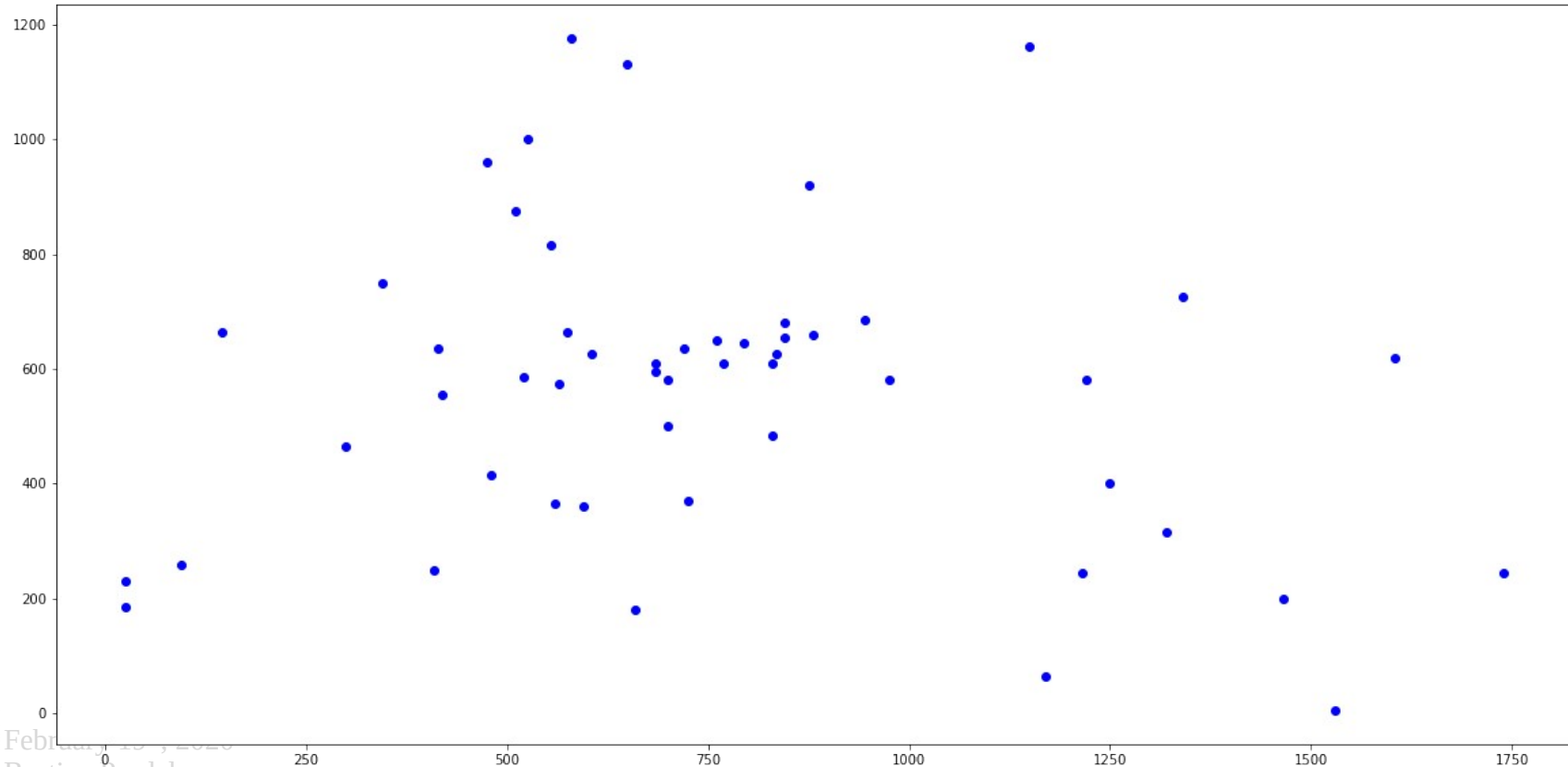
└ Obtained results



Results & analyses

└ Obtained results

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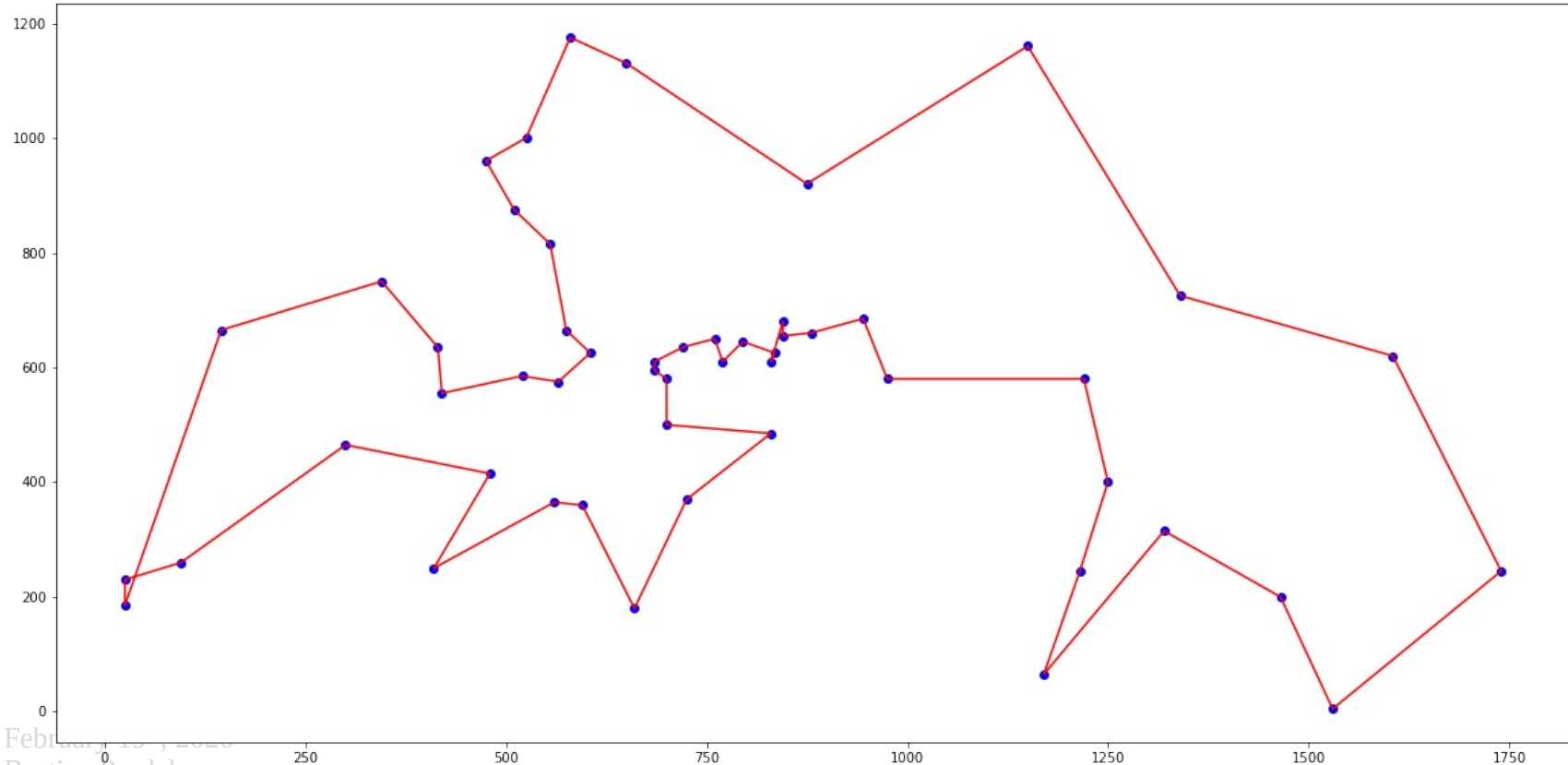


Results & analyses

└ Obtained results

Berlin52

Fitness: 0.9690350764486702

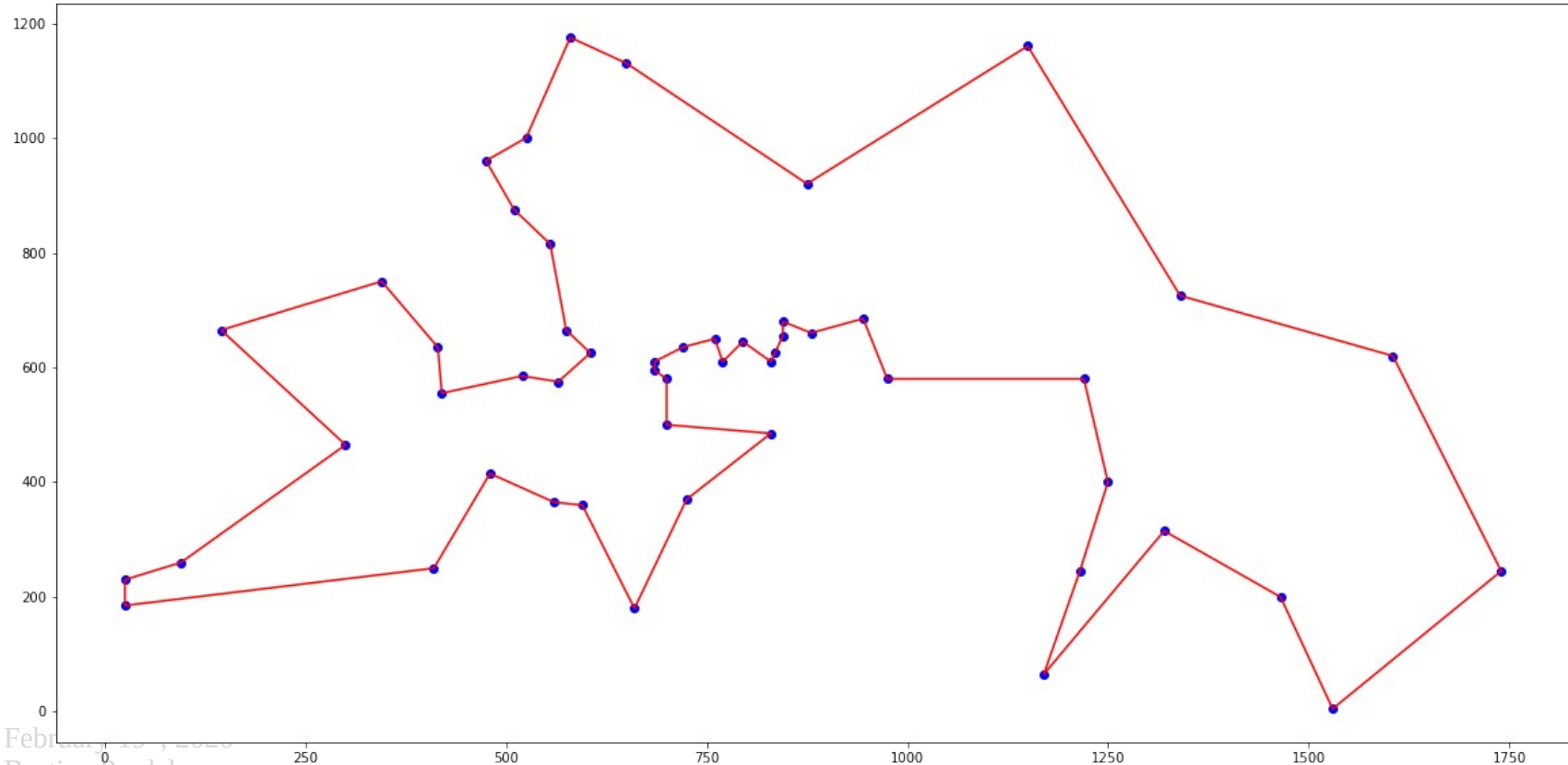


Results & analyses

└ Obtained results

Berlin52 (+ local search)

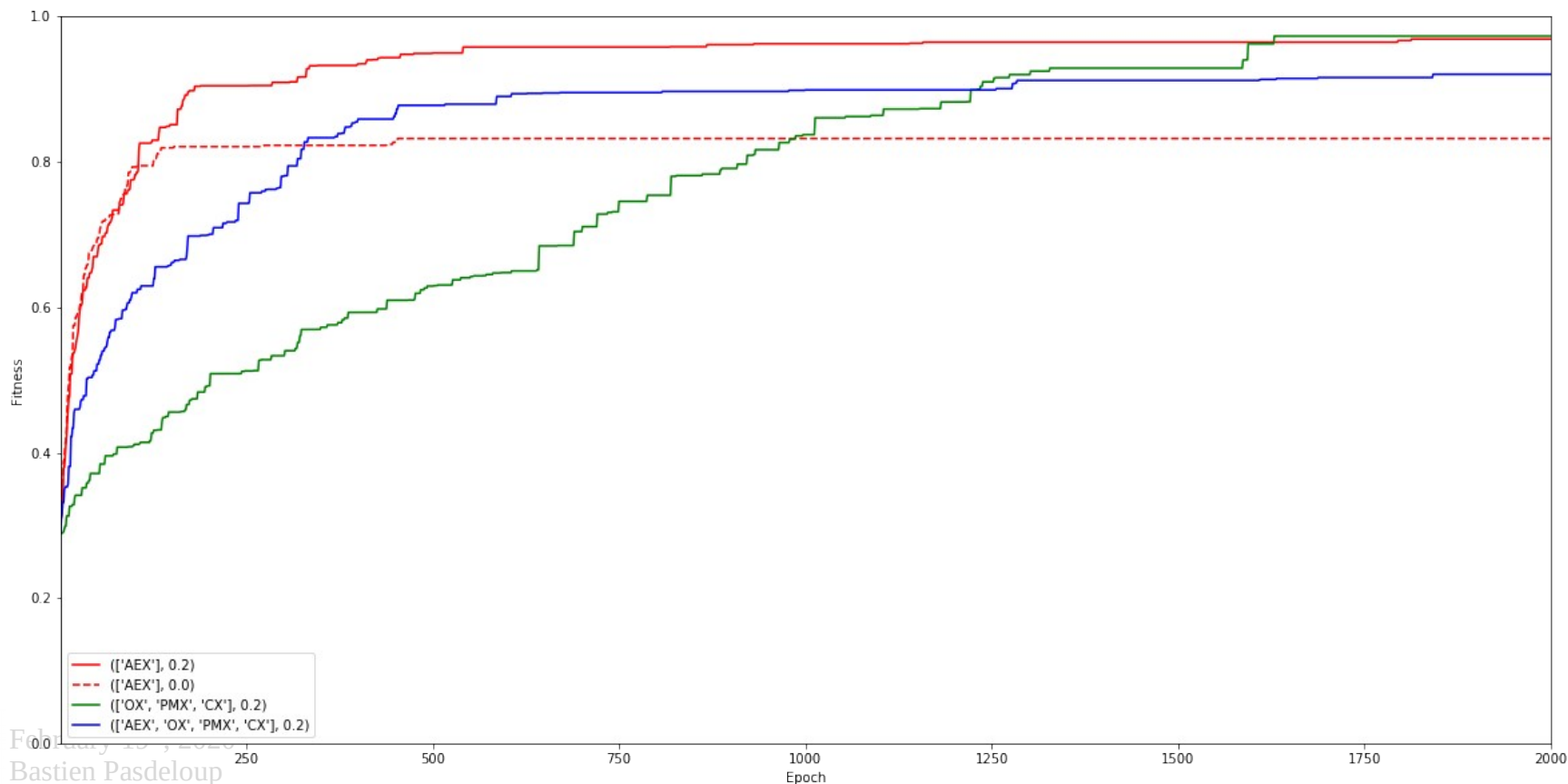
Fitness: 0.9900236282488842



Results & analyses

└ Comparison with other crossover operators (fixed number of epochs)

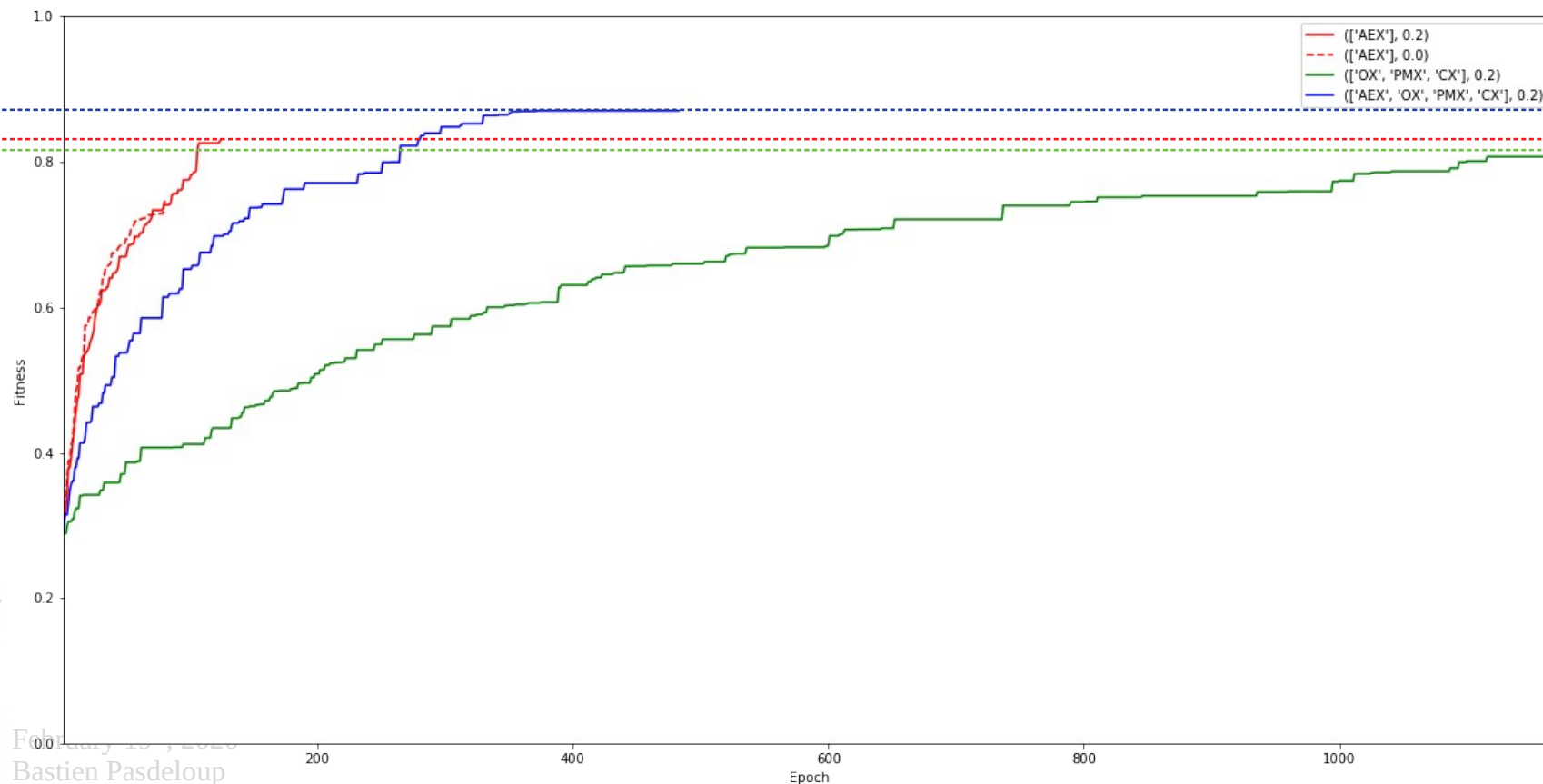
Berlin52



Results & analyses

└ Comparison with other crossover operators (stop after 1mn)

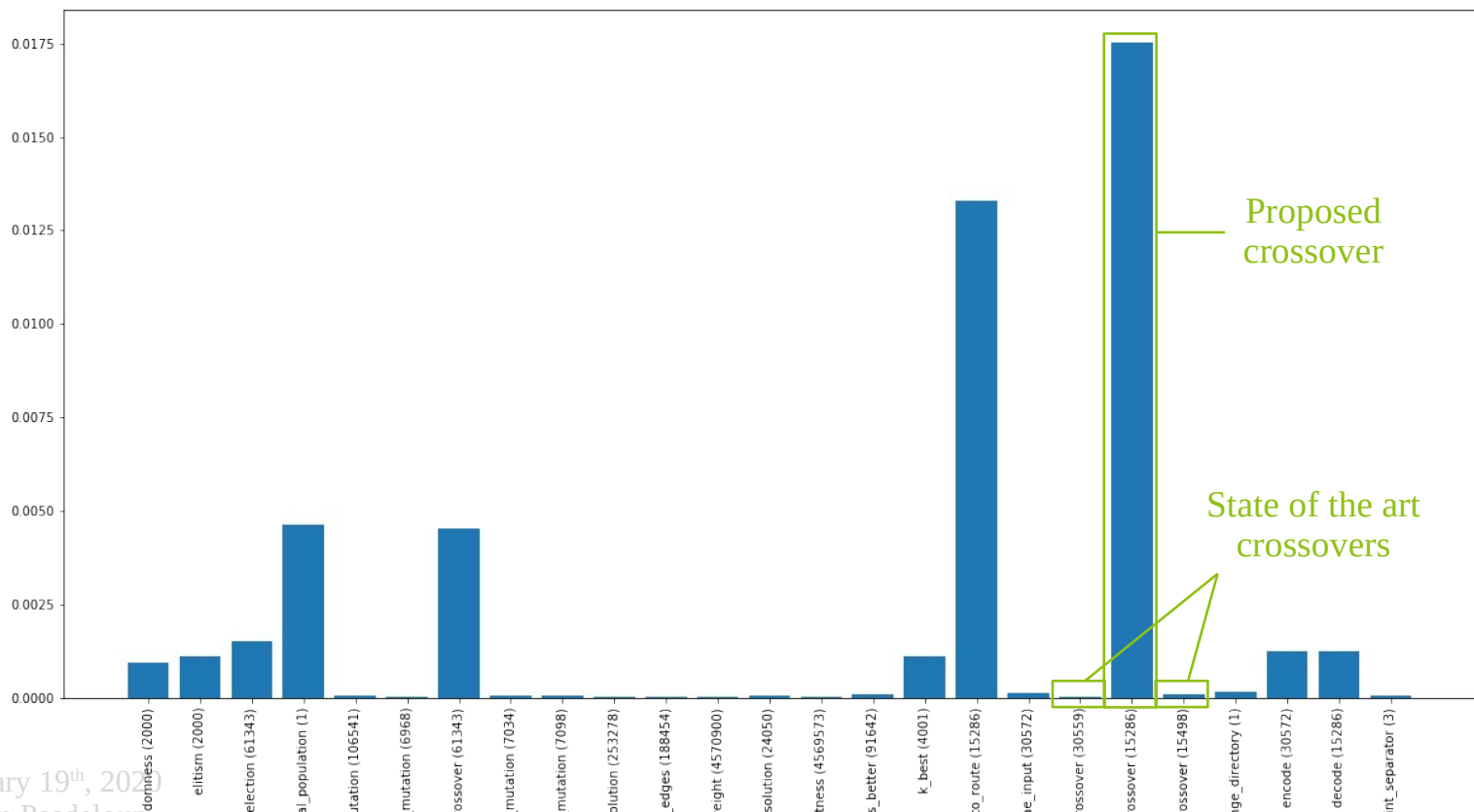
Berlin52



Results & analyses

└ Where is most of the time spent?

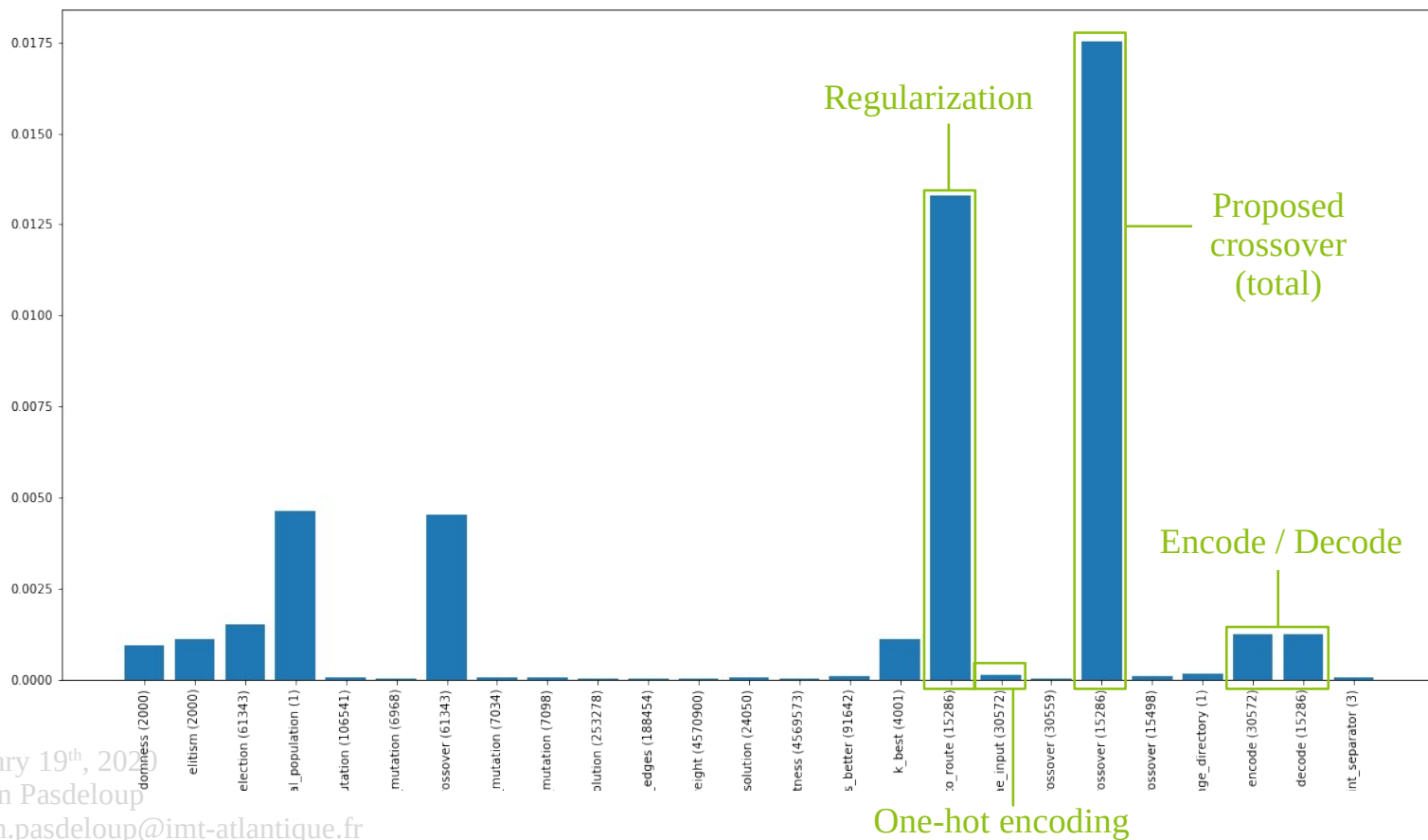
Berlin52



Results & analyses

└ Where is most of the time spent?

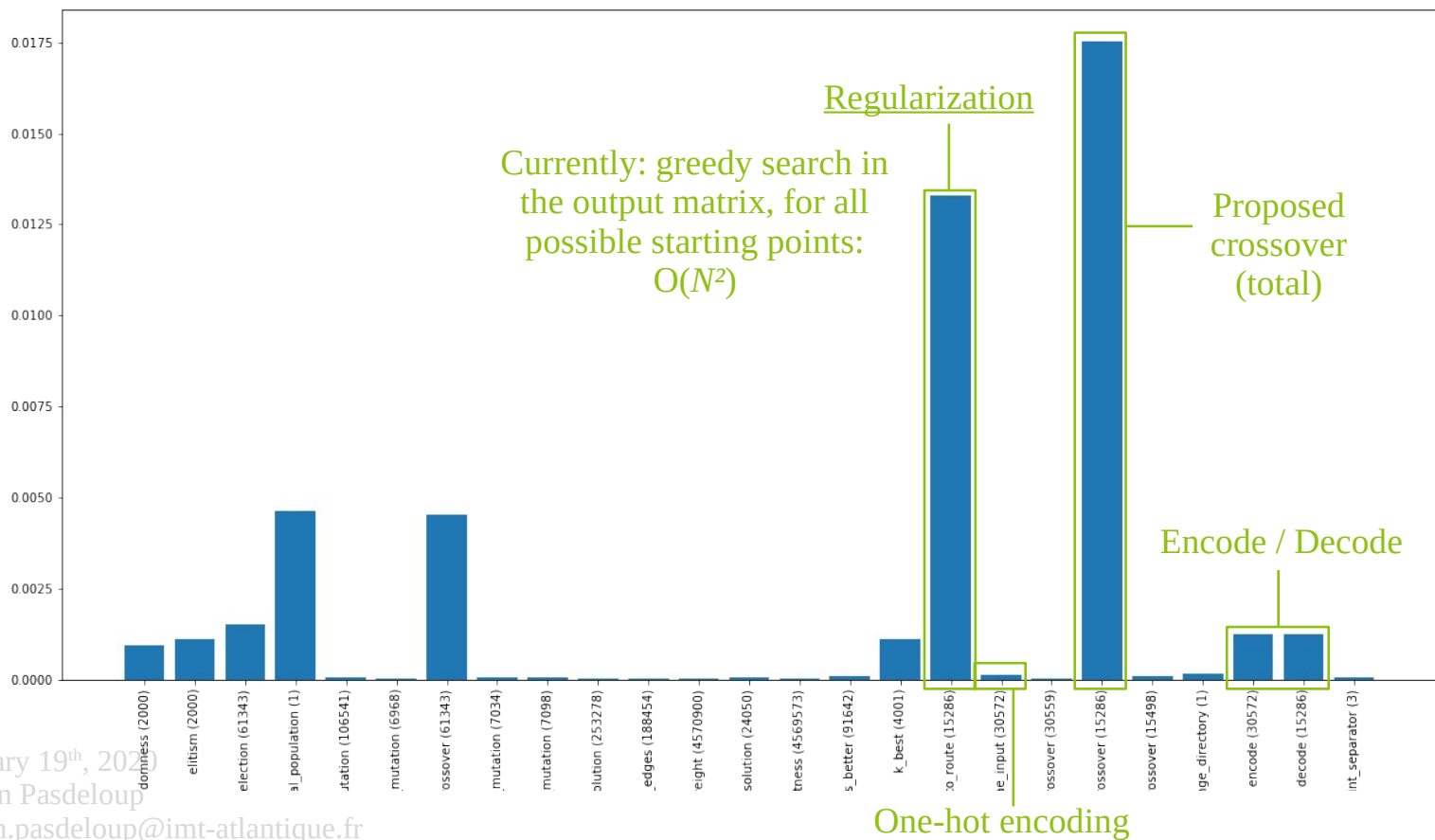
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Results & analyses

└ Where is most of the time spent?

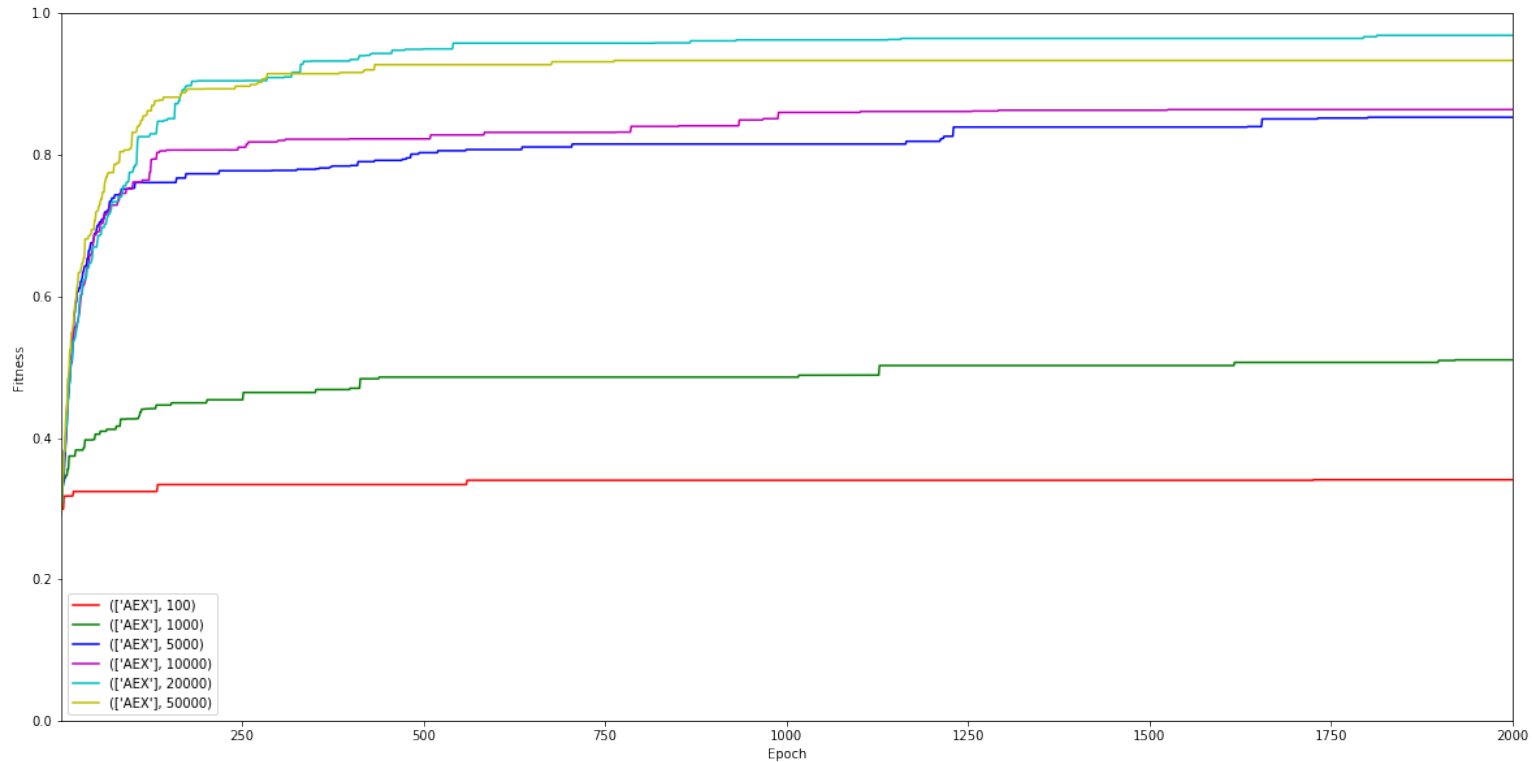
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Results & analyses

└ Individual variation of parameters (training set size)

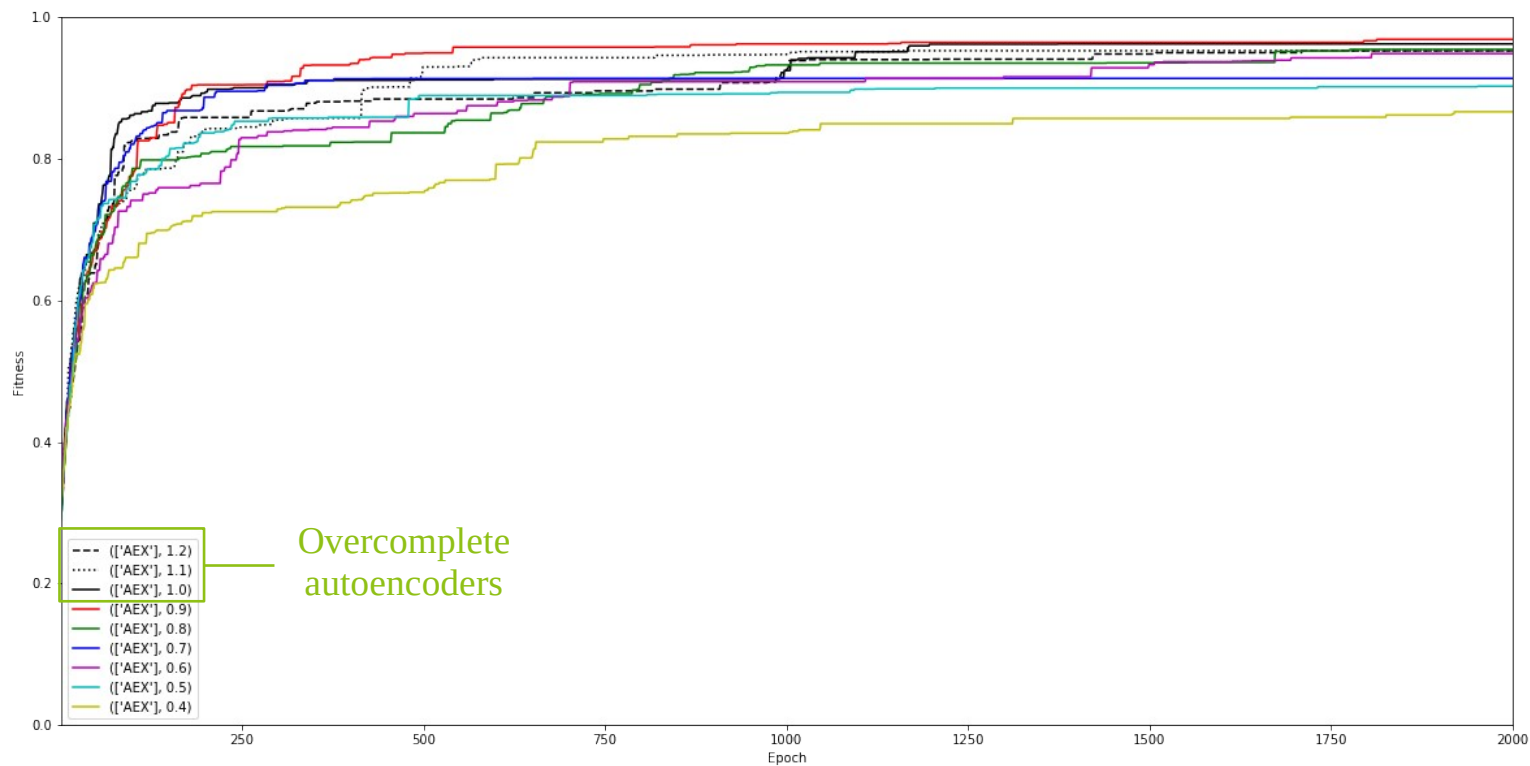
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Results & analyses

└ Individual variation of parameters (bottleneck dimension)

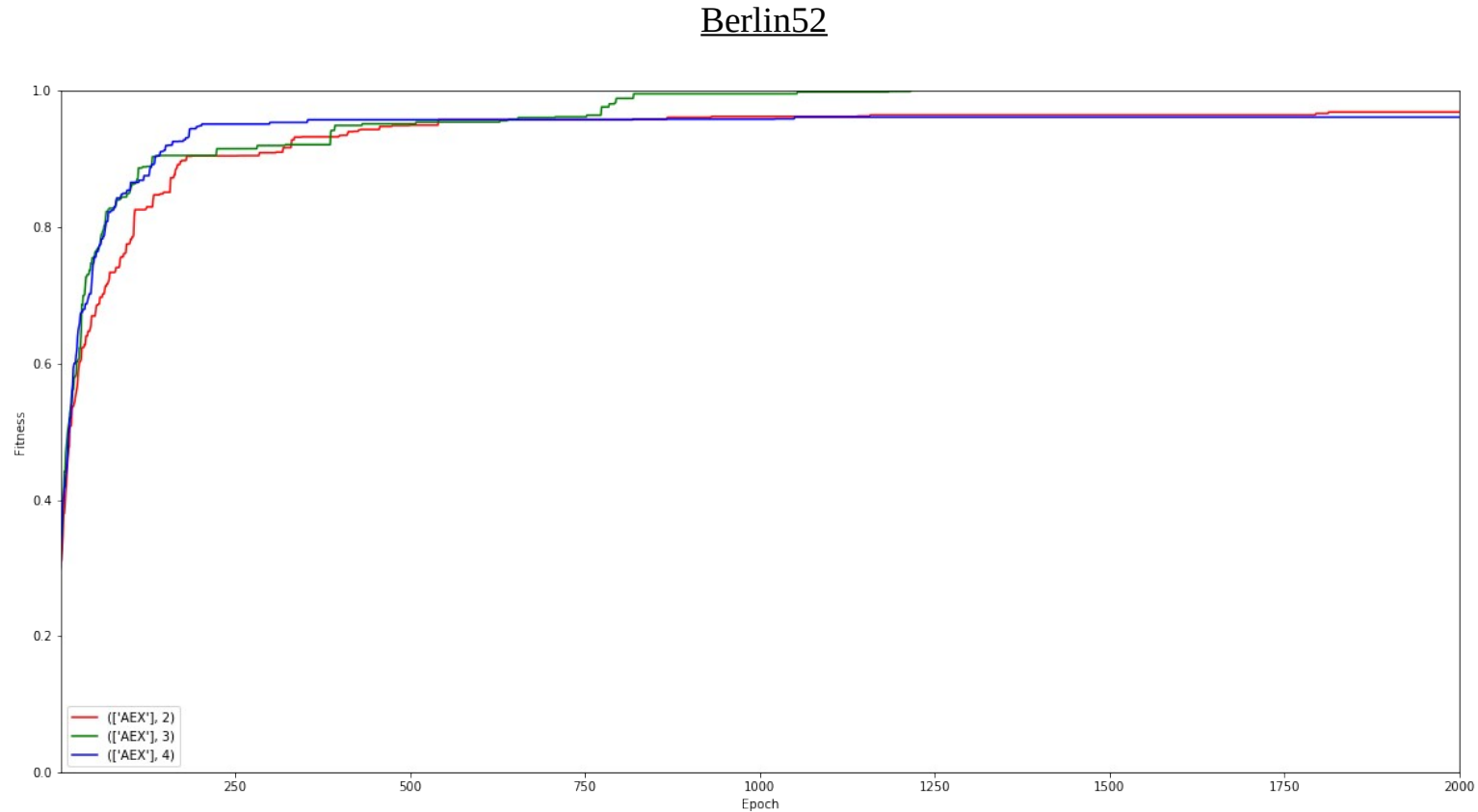
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Overcomplete
autoencoders

Results & analyses

└ Individual variation of parameters (number of parents per child)

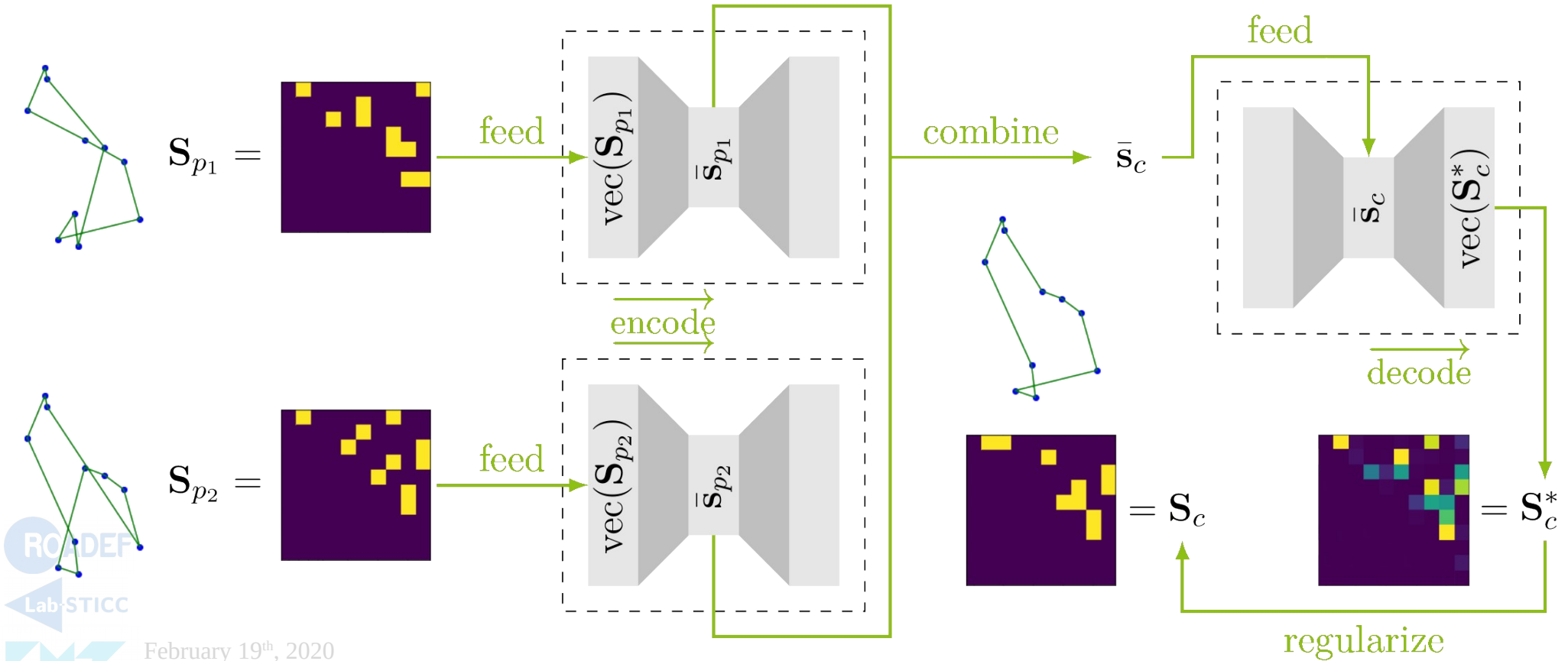


OUTLINE

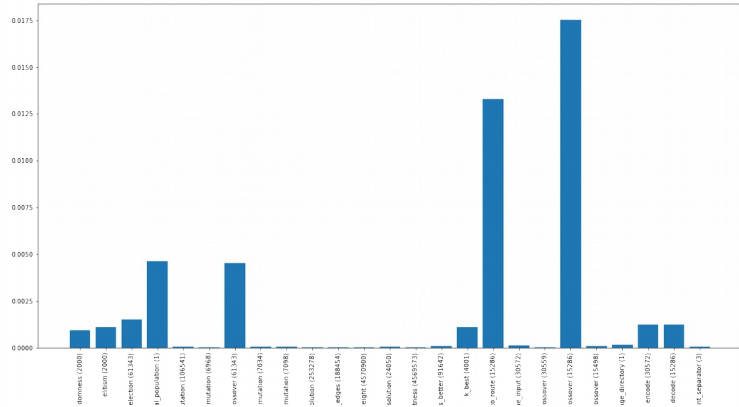
- ▶ Autoencoders & genetic algorithms
- ▶ Proposed operator
- ▶ Results & analyses
- ▶ **Conclusions & perspectives**

Conclusions & perspectives

Summary of the approach



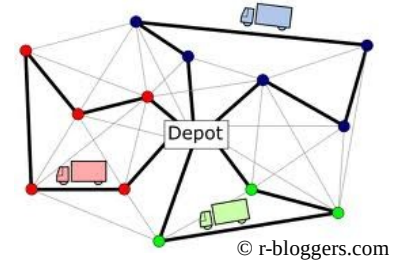
Perspectives



Make the approach faster
→ Another regularization?

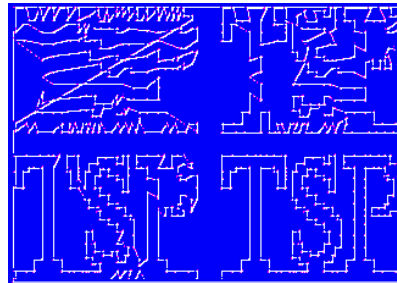


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Test applicability to other complex problems that are harder to solve than TSP



Tested on relatively small problems
→ Scale!!

Conclusions & perspectives

└ References

- [1] Geoffrey E Hinton and Richard S Zemel. Autoencoders, minimum description length and Helmholtz free energy. Advances in neural information processing systems, 3–10, 1994.
- [2] Yoshua Bengio, Grégoire Mesnil, Yann Dauphin and Salah Rifai. Better mixing via deep representations. International conference on machine learning, 552–560, 2013.
- [3] Christian Blum and Andrea Roli. Metaheuristics in combinatorial optimization : Overview and conceptual comparison. ACM Computing Surveys (CSUR), 35(3), 268–308, 2003.

[pdf] <https://easychair.org/publications/preprint/BQdg>

Autoencoder-based generation of individuals in population-based metaheuristics



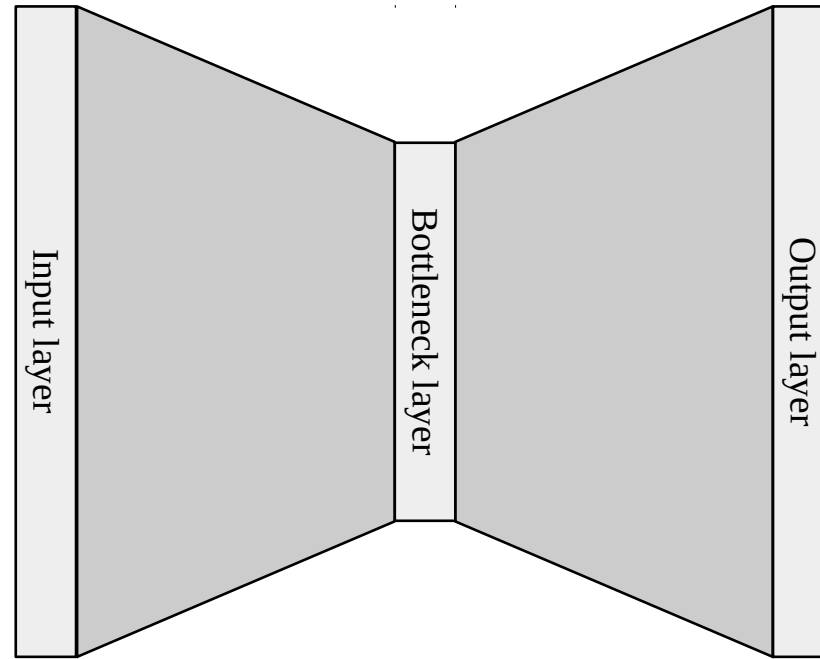
February 19th, 2020

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Supplementary materials

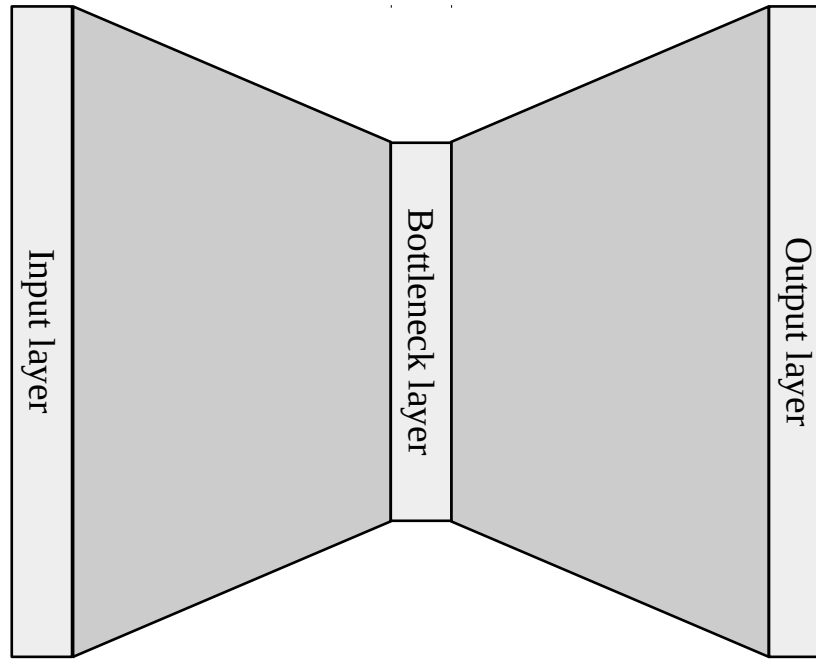
└ Other models based on autoencoders (denoising autoencoder)



Supplementary materials

└ Other models based on autoencoders (denoising autoencoder)

Corrupt images
during training

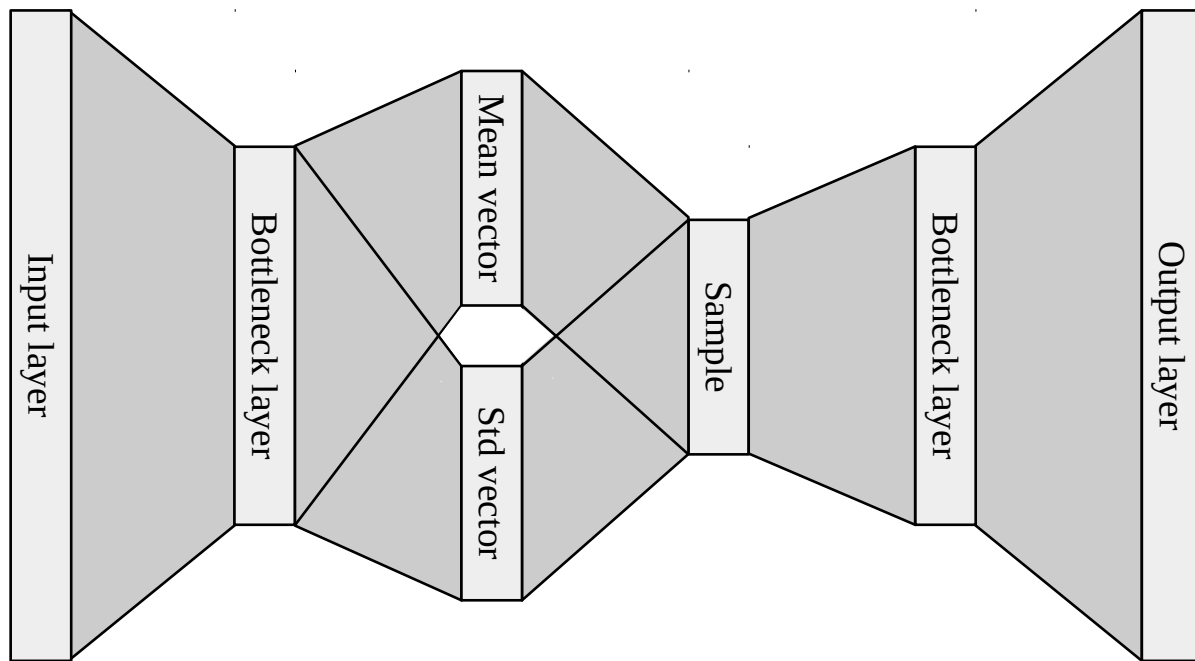


Target image is
non-corrupted version



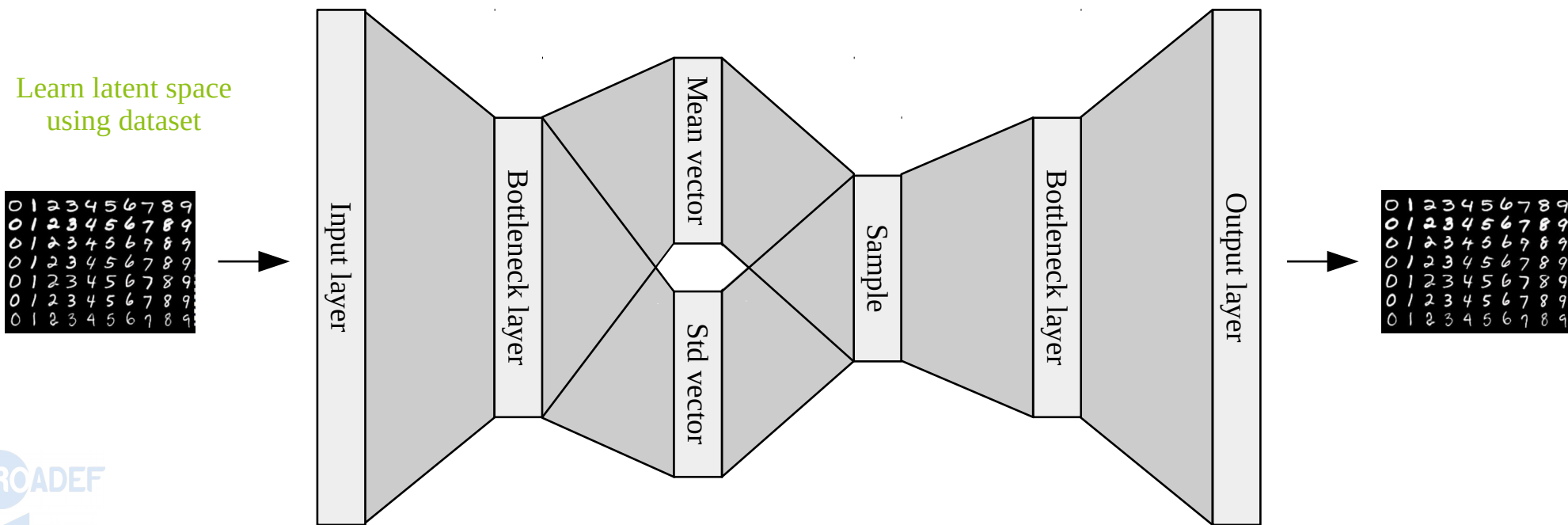
Supplementary materials

└ Other models based on autoencoders (variational autoencoder)



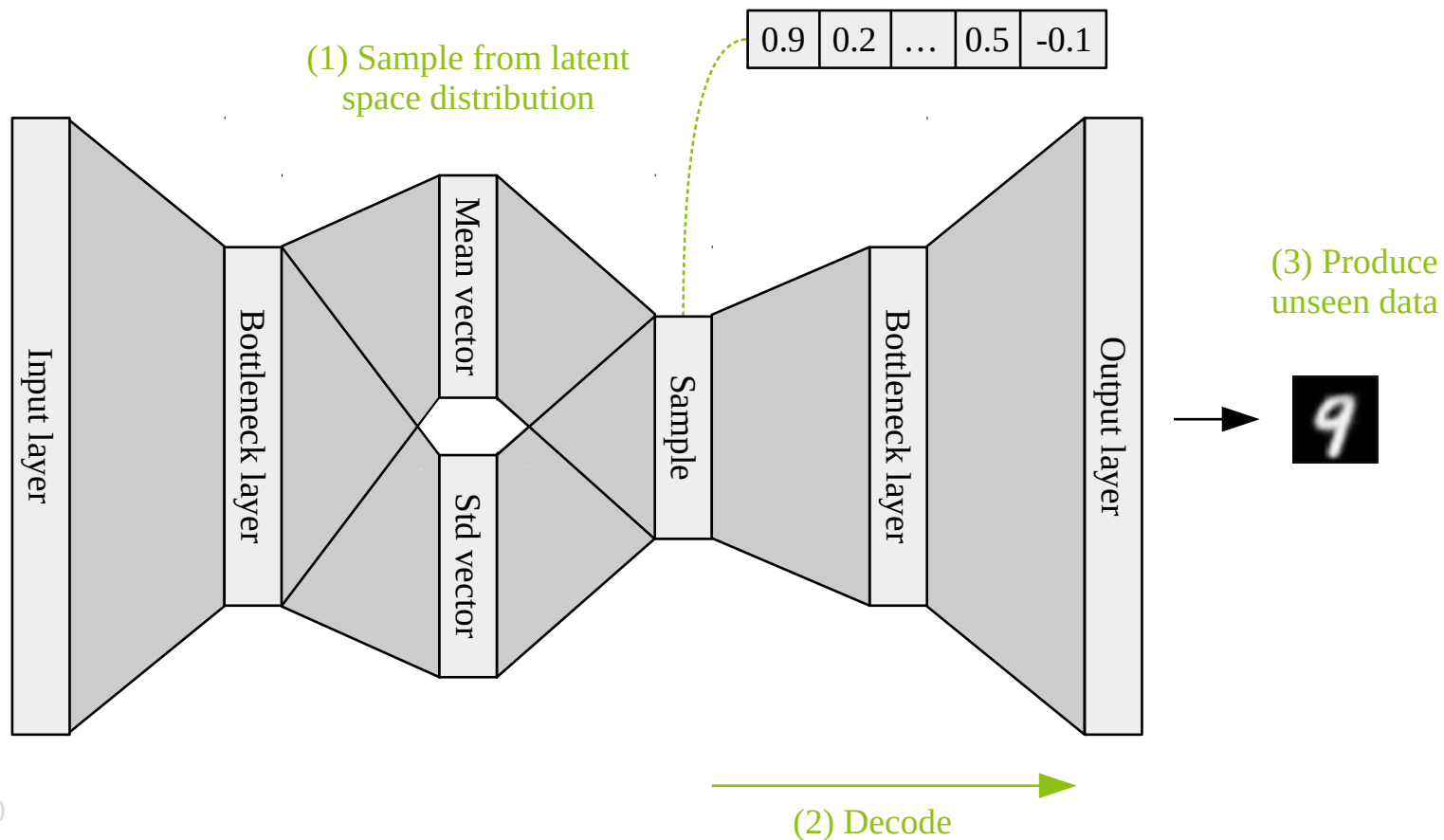
Supplementary materials

└ Other models based on autoencoders (variational autoencoder)



Supplementary materials

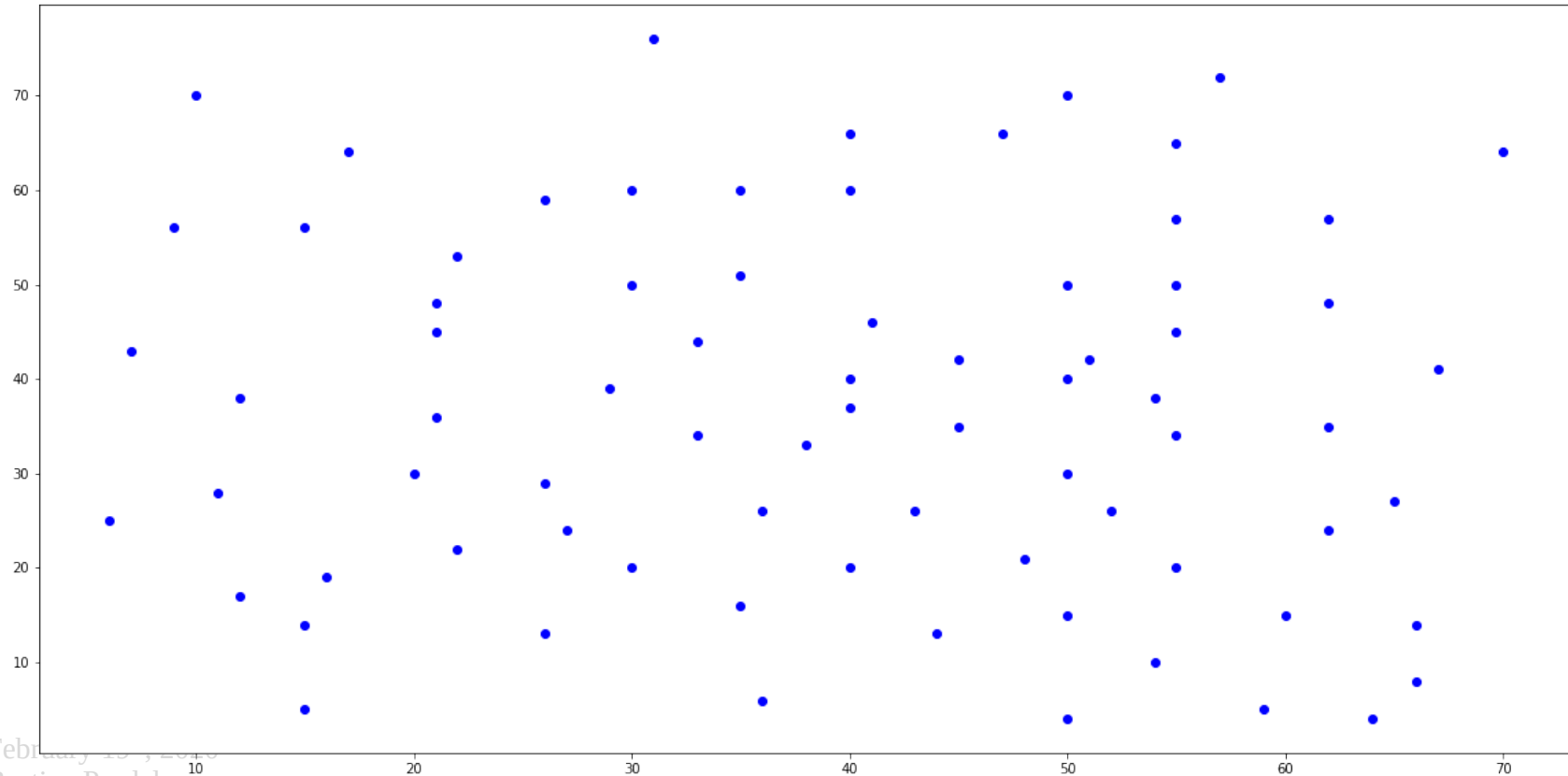
Other models based on autoencoders (variational autoencoder)



Supplementary materials

└ Obtained results

Eil76

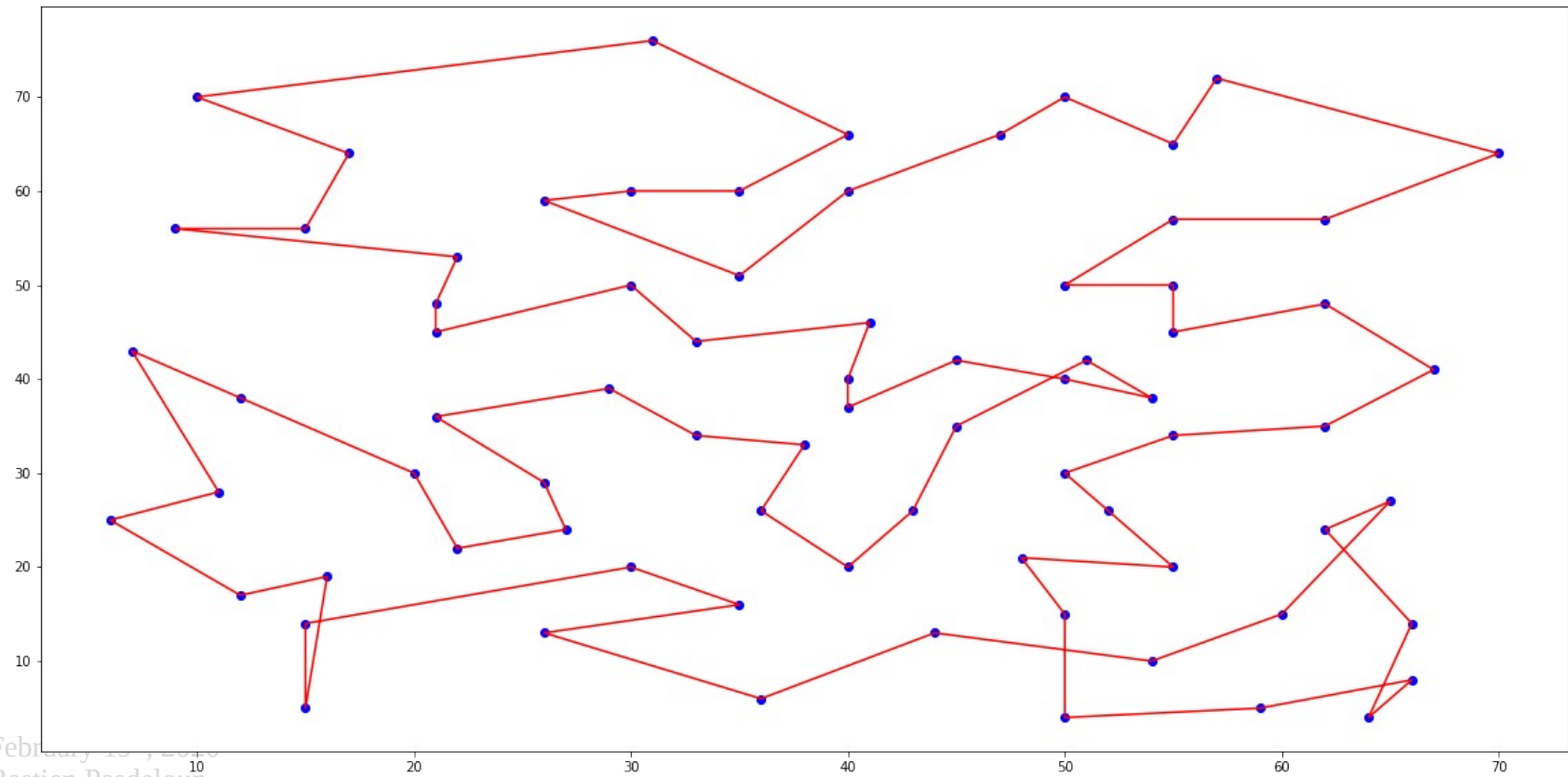


Supplementary materials

└ Obtained results

Eil76

Fitness: 0.867741935483871

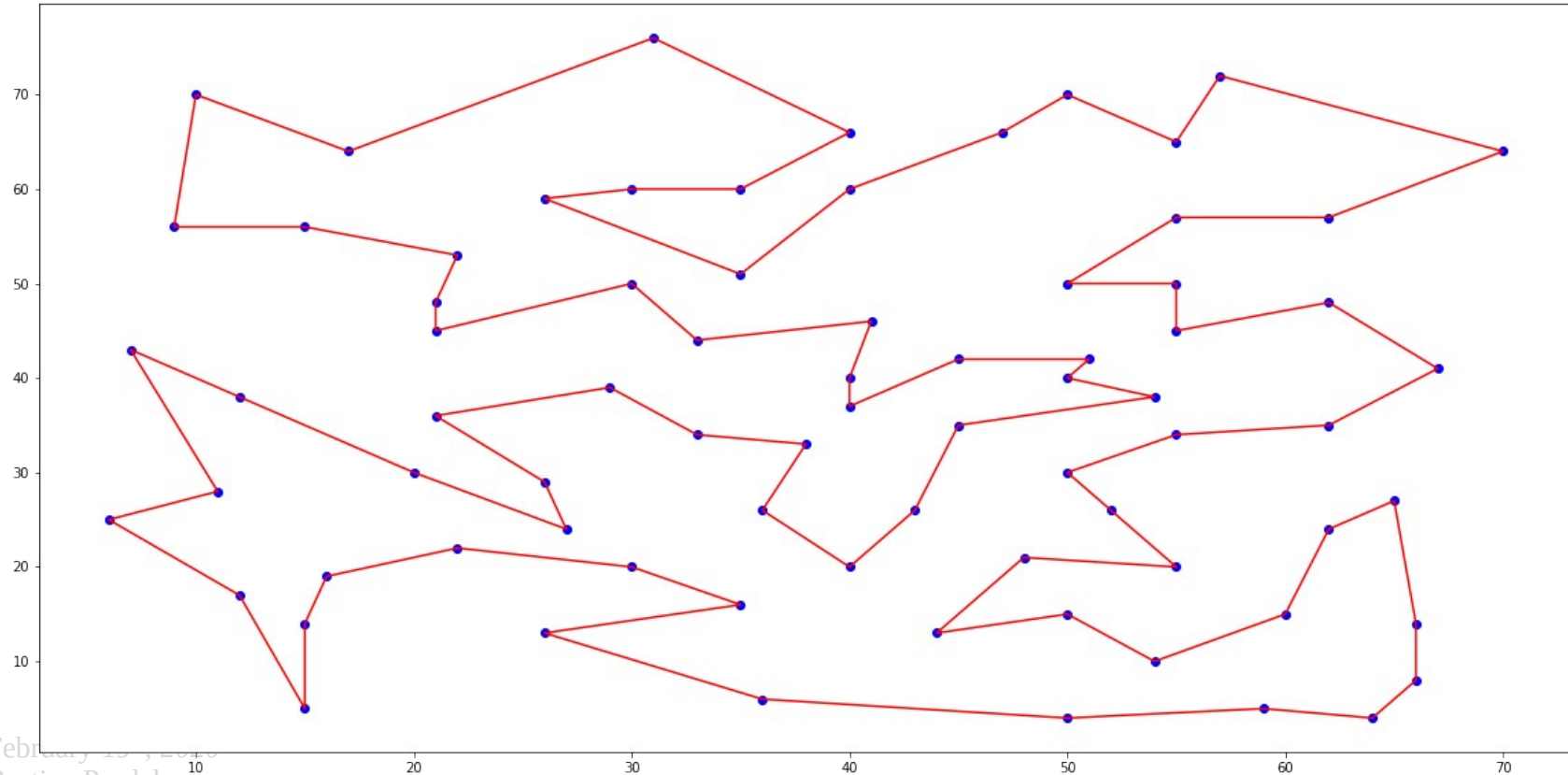


Supplementary materials

└ Obtained results

Eil76 (+ local search)

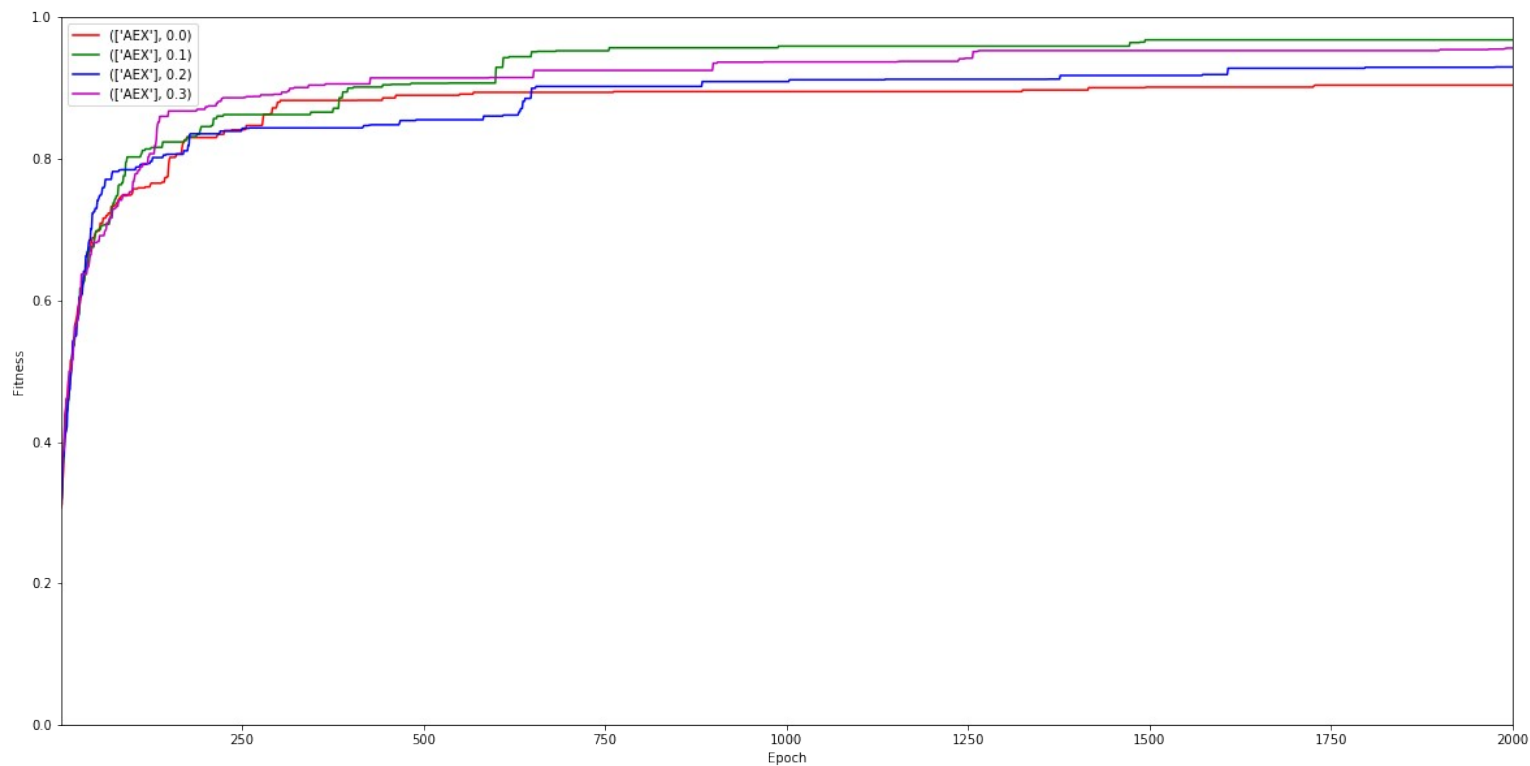
Fitness: 0.9011725293132329



Supplementary materials

└ Individual variation of parameters (random insertions)

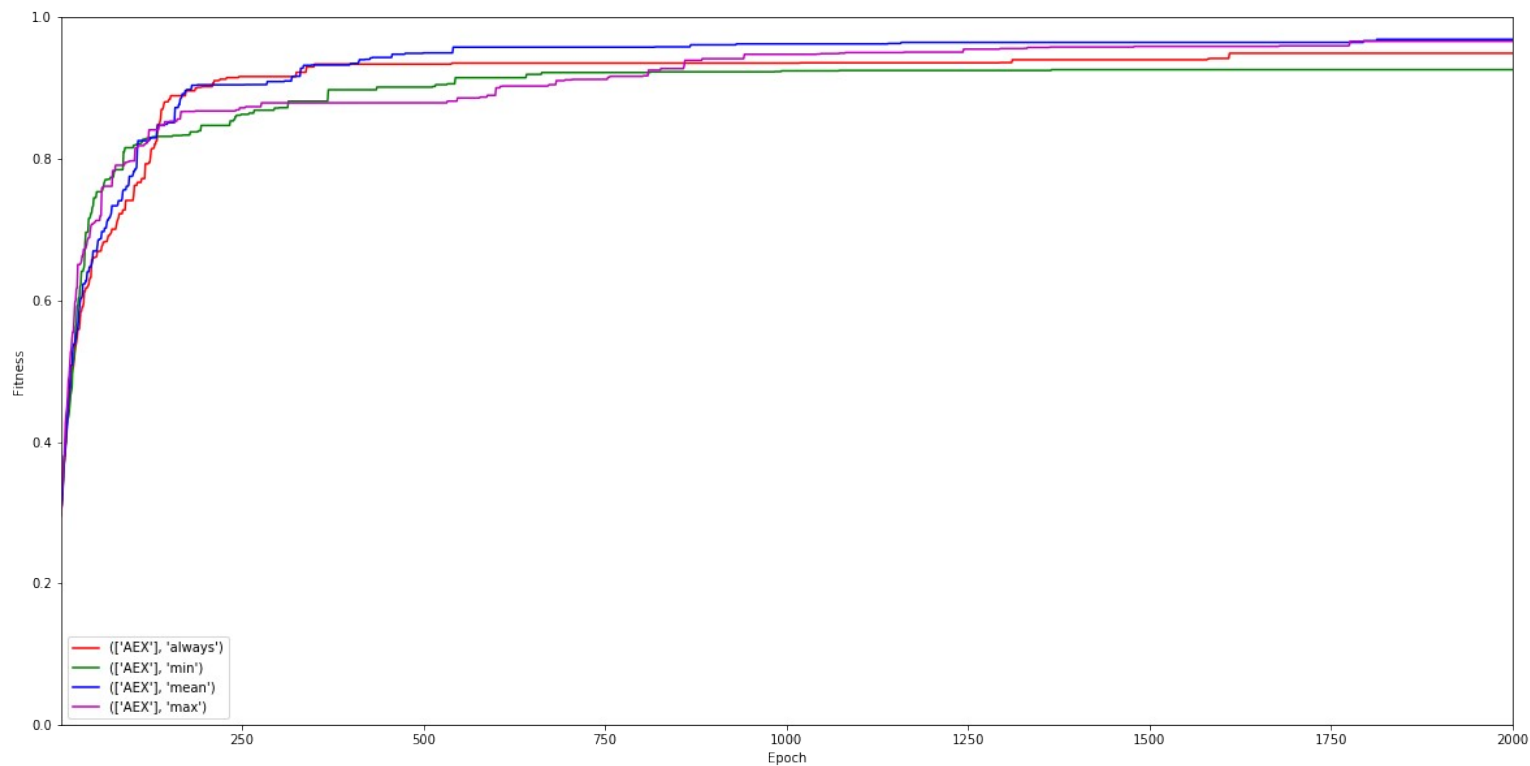
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Supplementary materials

└ Individual variation of parameters (children survival)

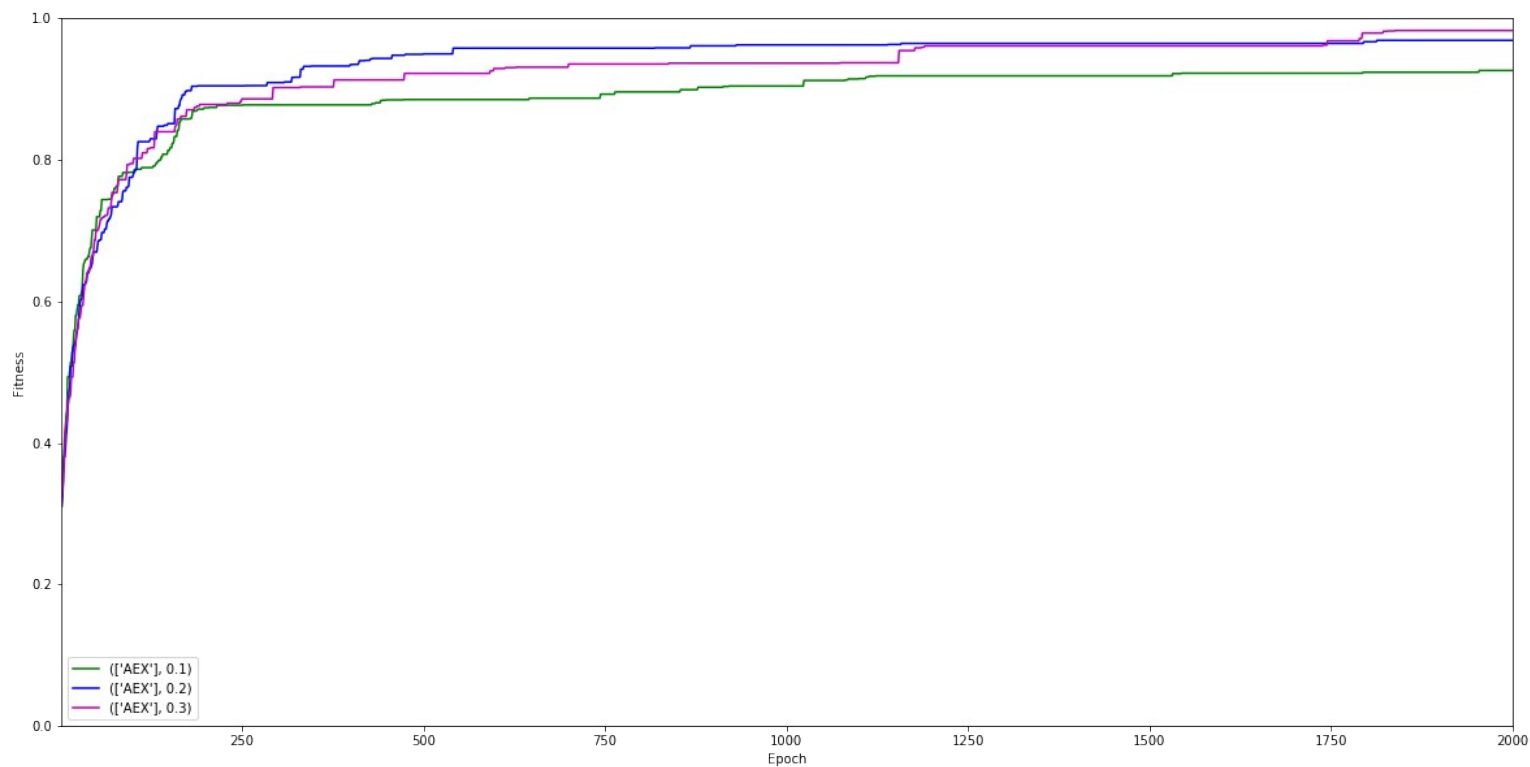
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Supplementary materials

└ Individual variation of parameters (elitism)

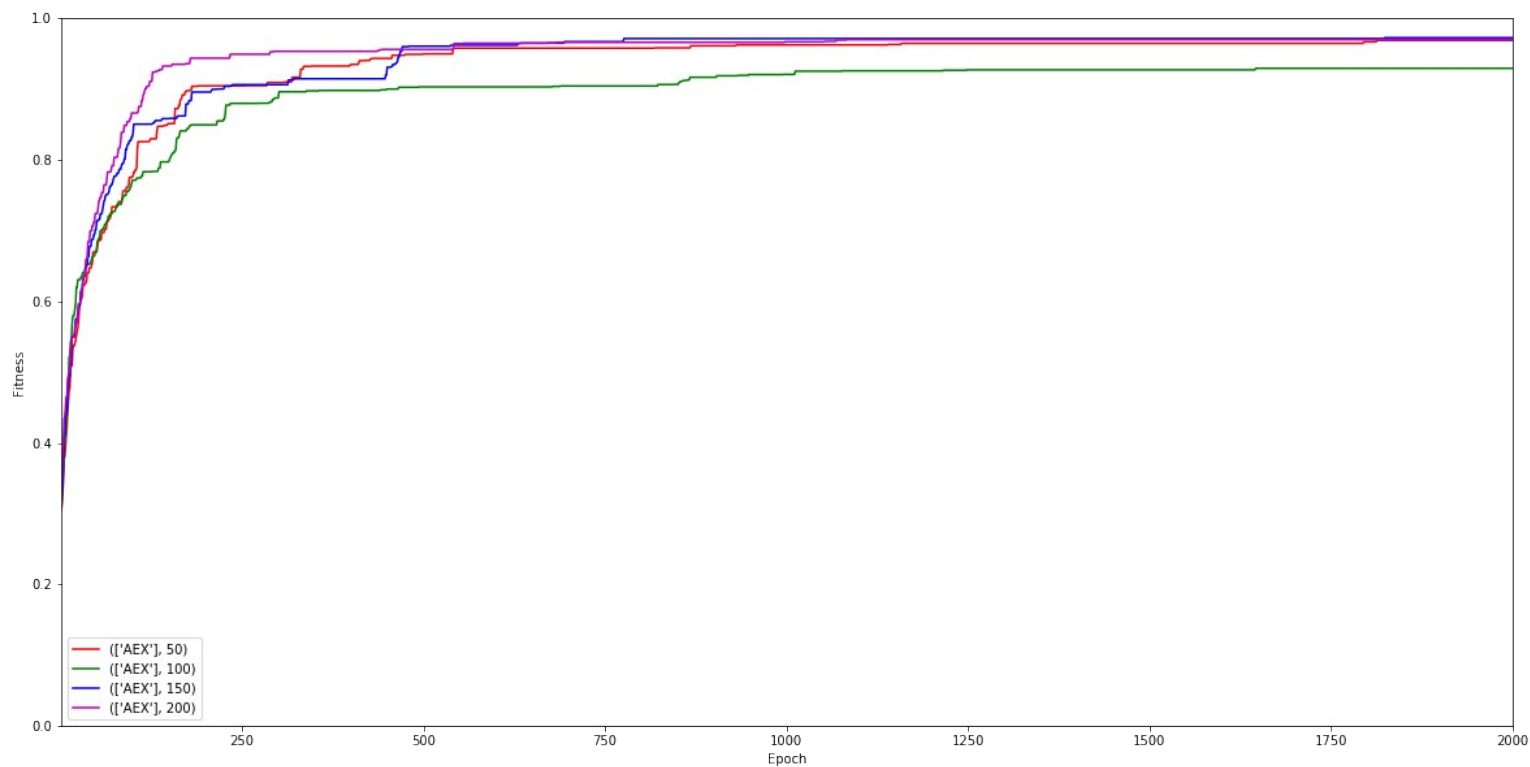
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Supplementary materials

└ Individual variation of parameters (population size)

Berlin52



Supplementary materials

└ Default parameters used

```
# Constants
DEBUG = True
OUTPUT_DIRECTORY = "../results/" + PROBLEM_NAME + "/experiment_1"
AUTOENCODERS_SAVE_DIRECTORY = "../trained_models"
AE_CONSTANTS = {"epochs" : 200,
                 "validation_ratio" : 0.9,
                 "batch_size" : 256,
                 "bottleneck_ratio" : 0.9}
GA_CONSTANTS = {"nb_parents_per_child" : 2,
                 "min_parents_distance" : 1,
                 "population_size" : 50,
                 "elitism_ratio" : 0.2,
                 "random_ratio" : 0.25,
                 "epochs" : 2000,
                 "timeout" : None,
                 "mutation_ratio" : 0.2,
                 "keep_child_better_than" : "mean",
                 "bad_child_survival" : 0.5,
                 "autoencoder_training_set_size" : 20000,
                 "crossovers" : ["AEX"],
                 "mutations" : ["displacement", "double_bridge", "reverse_sequence"],
                 "local_searches" : ["2-opt"],
                 "local_search_when" : []} # No local searches

# Run experiment
best_fitness_per_epoch, best_solution = go()
```